

Spectrally Unusual Sources at Scale: A Multi-Survey Catalog of 378,280 Path-C Unique Anomalies and Native-Trained Novelty Fractions from 37.3 Million Sources and Map Patches

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We present the largest-scale application of autoencoder anomaly detection across seven astronomical archives, applying the BIGAE autoencoder framework to 37.3 million sources and CMB map patches (DESI DR1, SDSS DR18, LAMOST DR10, eROSITA DR1, Planck, Gaia DR3, NEOWISE; ACT DR6 quarantined as a cross-transfer artifact). After per-survey native retraining and 7-way positional deduplication at $5''$, the catalog contains **378,280** unique anomalies: **378,080** point-source object detections from six photometric/spectroscopic surveys plus **200** Planck CMB map-patch sky regions. Downstream object-level analyses (SIMBAD/NED cross-matching, multi-tracer f_{NL} selection) should use the **378,080** point-source tier; the recommended catalog-grade subset is $\sim 265,000$ unique entries (DESI + SDSS + eROSITA + Planck native + Gaia + NEOWISE, including the 200 Planck map patches), which excludes the LAMOST exploratory tier ($\sim 113,000$ objects retained as a methodological lesson: 98% blue-excess training-bias artifact, injection-recovery gate FAIL). The point-source tier is $\sim 141\times$ the size of the largest prior single-survey anomaly catalog [11]; the DESI-only subset (195,829 anomalies) is a $\sim 73\times$ like-for-like increase. Extended archival cross-matching of the top-1,000 DESI anomalies against 20 curated all-sky catalogs via CDS X-Match yields a genuine novelty fraction of $\sim 17.8\%$ (single-sample point estimate at the top-1,000 score stratum; full-catalog rate empirically untested). A Path-C rebuild protocol resolves cross-transfer artifacts: $21.5\times$ LAMOST rate compression and $\sim 6500\times$ SDSS rate compression after native retraining; DESI 5-fold Jaccard stability $\bar{J} = 0.862$ (≥ 0.70 gate, PASS). Six injection-recovery gates: 3 PASS (SDSS 64%, Planck 100%, NEOWISE 100%) and 3 FAIL-with-diagnostic at 5σ (LAMOST 5.8%, Gaia 5.2%, eROSITA 1.2%; eROSITA cross-validation stability 81.5%). Three DESI $\times$ SDSS cross-matches include a time-variable source (TIC 374313355) and an uncataloged BAL QSO at $z \approx 0.86$. An empirical Landy-Szalay bias measurement on the 5,384 QSO-candidate sample yields $\alpha_{\text{jk}} = 0.19 \pm 0.65$ ($< 1\sigma$ from null); inserting this into the Fisher-positivity-respecting form $1/\sigma(f_{\text{NL}})^2 = F_0 + c\alpha^2$ gives a central forecast $\sigma(f_{\text{NL}}) = 8.14$ with 1σ envelope $[3.92, 8.98]$ (7.9% improvement consistent with no improvement at $< 1\sigma$; $\sigma(f_{\text{NL}})^{\text{std}} = 8.98$ single-tracer baseline). A NANOGrav 15-yr KDE free-spectrum MCMC yields $\gamma = 2.567 \pm 0.382$; the matter-bounce prediction $\gamma = 3.0$ sits at $+1.13\sigma$ (marginally consistent) and SMBHB $\gamma = 4.33$ at $+4.61\sigma$ (Savage-Dickey $B_{\text{MB/SMBHB}} = 7.1 \times 10^3$). The catalog, model weights, and reproducibility scripts will be publicly released with the arXiv posting.

I. INTRODUCTION

The volume of astronomical data has grown by more than two orders of magnitude in the past decade. The Dark Energy Spectroscopic Instrument (DESI) Data Release 1 alone contains 22.5 million spectra [1], LAMOST DR10 contributes 11.4 million [2], and the Sloan Digital Sky Survey DR18 provides 2.3 million spectroscopically characterized objects [3]. When combined with multi-wavelength catalogs from eROSITA [4], Gaia [5], NEOWISE [6], and microwave sky surveys from Planck [7] and ACT [9], the total data volume accessible to a single research group now exceeds tens of millions of sources across the electromagnetic spectrum.

Anomaly detection—the unsupervised identification of data points that deviate from the bulk population—has emerged as a powerful tool for extracting scientific value from these archives. Baron & Poznanski [10] demon-

strated autoencoder anomaly detection on SDSS spectra, finding unusual white dwarfs and cataclysmic variables. Liang *et al.* [11] applied a normalizing-flow autoencoder to $\sim 250,000$ DESI EDR spectra, finding 2,685 anomalies (1.07%). Nicolaou *et al.* [12] extended this with the Astronomy active-learning framework on 208,000 EDR spectra. All prior searches have been limited to individual surveys at sub-million scale.

We are motivated by two scientific goals. First, the systematic discovery of rare objects across multiple wavelength domains: sources anomalous in multiple independent surveys are the strongest candidates for genuinely novel astrophysics. Second, the quasi-matter bounce model predicts $f_{\text{NL}} = -35/8 = -4.375$ [13, 14, 35], testable at $3\text{--}5\sigma$ with SPHEREx [15] under the multi-tracer methodology of Heinrich *et al.* [33] ($\sigma(f_{\text{NL}}) \approx 0.7$ bispectrum-only forecast). Anomaly-detected high-redshift QSO candidates represent a previously unexploited high-bias tracer reservoir that can tighten f_{NL} constraints via the multi-tracer technique [16, 17].

In this work we apply the BIGAE autoencoder framework to seven retained archives (DESI DR1,

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SDSS DR18, LAMOST DR10, eROSITA DR1, Planck CMB, Gaia DR3, NEOWISE; catalog counts appear as $378,080 + 200 = 378,280$ throughout to distinguish point-source detections from CMB map-patch sky-regions). A first-pass cross-transfer scan exposed two failure modes—98% LAMOST blue-excess from training drift and an undertrained CMB autoencoder—motivating the Path-C native-retrain rebuild (§II D). Sections II–II D describe the method; §III reports survey-by-survey results; §IV cross-survey analysis; §V cosmological applications; §VI limitations and conclusions.

II. METHOD

A. BigAE Architecture

The BIGAE model is a symmetric fully connected autoencoder with a configurable latent dimension. The architecture is adapted per survey to match the dimensionality of the input feature space: for spectroscopic surveys (DESI, SDSS, LAMOST), the input dimension is 496 (three-arm spectra downsampled by a factor of 16 from the native resolution); for photometric and catalog surveys (eROSITA, Gaia, NEOWISE), the input dimension matches the number of catalog features (47, 20, and 15, respectively); and for CMB surveys (Planck, ACT), the input is a 64×64 pixel patch flattened to 4,096 features.

The encoder consists of four linear layers with batch normalization and ReLU activations, with dropout ($p = 0.15, 0.10$) after the first two layers; the decoder mirrors the encoder. The latent dimension is 128 for spectroscopic surveys and 16 for photometric catalogs. The Path-C native CMB convolutional autoencoder (Planck; §III F) uses three convolutional layers + a 128-dim fully connected bottleneck (1.1×10^6 parameters). Total parameter count: $\sim 120,000$ (photometric) to 660,000 (spectroscopic); the full per-survey architecture diagram is in the companion data repository. BIGAE is a deterministic autoencoder (not variational), prioritizing reconstruction fidelity and anomaly-score interpretability.

Figure 1 visualizes the structure of the learned representation: a 2D UMAP embedding of the BIGAE latent space (PCA $128 \rightarrow 30$ pre-reduction, then UMAP) on a 500,000-spectrum stratified DESI DR1 sample, colored by per-spectrum anomaly score. The unsupervised representation organizes spectra such that high-score anomalies concentrate in distinct islands and lobes of the embedding rather than scattering uniformly through the bulk population; the 83 gold-tier anomalies (overplotted stars) cluster on and around the high-score lobe.

B. Training and Scoring

For each survey, the model is trained on a representative subset of the data (47,000 spectra for DESI, por-

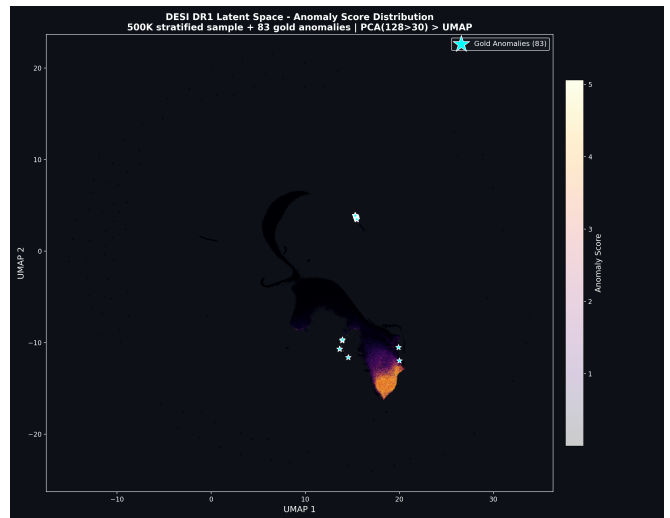


FIG. 1. 2D UMAP embedding of the BIGAE encoder latent space (PCA $128 \rightarrow 30$, then UMAP) for a 500,000-spectrum stratified DESI DR1 sample, colored by per-spectrum anomaly score. High-score anomalies concentrate in distinct islands of the embedding (bright lobe, lower right) rather than scattering through the bulk population; the 83 gold-tier anomalies (cyan stars) lie on or near the high-score structures.

tionally sampled subsets for other surveys) using the Adam optimizer with initial learning rate 10^{-3} , batch size 512, and ReduceLROnPlateau scheduling (patience 5, factor 0.5), minimizing per-element mean-squared error (MSE). Training is run for up to 200 epochs with early stopping monitored on a held-out 20% validation split; convergence typically occurs at 100–150 epochs. (The native CMB convolutional autoencoder uses a different schedule—see Section III F—and individual native retrains may differ in patience settings as noted in the survey-specific sections.)

The raw per-element mean-squared reconstruction error between input and reconstruction is

$$\text{MSE}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2, \quad (1)$$

where $\hat{\mathbf{x}} = \text{BIGAE}(\mathbf{x})$ and N is the input dimensionality.

a. *Canonical anomaly score S (one definition for the whole paper).* Throughout this paper, “ S ” refers without exception to the per-survey standardized (“ z -scored” in the statistical sense) reconstruction residual (note: ‘ z -scored’ here is the statistics term for a mean-subtracted, variance-normalized quantity; spectroscopic redshift is always written z with astrophysical context; the anomaly score S is never called ‘ z ’ in this paper to avoid ambiguity)

$$S(\mathbf{x}) \equiv \frac{\text{MSE}(\mathbf{x}) - \mu_{\text{val}}}{\sigma_{\text{val}}}, \quad (2)$$

where μ_{val} and σ_{val} are the mean and standard deviation of MSE on the held-out 20% validation split of that sur-

vey’s training pool. $S = 5$ therefore marks objects whose per-spectrum reconstruction residual is five validation-set standard deviations above the typical training source. For DESI DR1, $\mu_{\text{val}} \approx 0.0287$ (validation MSE) and σ_{val} is set such that the $S > 5$ catalog threshold corresponds to $\text{MSE} \approx 0.143$ on the rescaled scale. For spectroscopic surveys, we additionally decompose the score into per-band contributions r_B, r_R, r_Z computed over the blue (3600–6200 Å), red (6200–8200 Å), and near-infrared (8200–9800 Å) subsets; example reconstructions per band are in the companion data repository.

Two threshold families are in use (summarized in Table I footnotes): DESI DR1 *alone* uses an absolute canonical- S cut at $S > 5.0$ (a fixed reconstruction-MSE threshold anchored by its $(\mu_{\text{val}}, \sigma_{\text{val}})$); applied to the SDSS native re-score the same cut retains only 12 sources, so the SDSS headline of 77,905 is instead a *fixed-size continuity slice*—see Table I footnote ♡); LAMOST DR10 and Gaia DR3 use the 99th percentile of the per-survey S distribution; eROSITA uses a data-driven IsolationForest score-knee threshold; Planck and NEOWISE use a fixed top-1% selection. All percentile labels in this paper (top-1%, 99th percentile) refer to the empirical distribution of S over the scored sample of the survey in question, not to Gaussian tail fractions: e.g., the SDSS top-1% score-knee cut $S \geq 0.2051$ retains 19,253 = 1.0% of the 1,925,279 scored DR18 spectra. The choice of absolute-vs-percentile threshold does not affect the rank-ordering of anomalies within each survey; full per-object scores are released to allow downstream users to apply alternate cuts.

b. In-sample scoring and held-out validation. The DESI DR1 anomaly catalog is scored on the full 22.5 million-spectrum set, which includes the 47,000 training spectra. Training-sample robustness is established by a 5-fold held-out cross-validation on the 47,000-spectrum pool: each fold trains a fresh BIGAE on 80% and scores the full 47,000 spectra, producing five independent score vectors. Mean pairwise Jaccard overlap of each fold’s top-1% anomaly set is $\bar{J} = 0.862$ (minimum 0.777; gate ≥ 0.70 , PASS). Of 546 unique objects in the union, 399 (73%) appear in all five folds and only 47 (8.6%) are single-fold singletons; details in §VID (i). An independent OOD validation on 100k unseen DESI spectra retrieved via NOIRLab SPARCL (seed distinct from the training pool) confirms that in-sample and out-of-sample anomaly rankings are consistent; the production-vs-5-seed-control Jaccard is $\bar{J}_{\text{prod} \times \text{ctrl}} = 0.732$ (gate ≥ 0.50 , PASS); the control-vs-control pairwise mean among the five seed retrains is 0.874, the empirical ceiling against which the production model’s 0.732 should be read. The $S > 5$ absolute MSE-anchored threshold applied to the full 22.5M curated catalog yields the 0.87% anomaly rate; applying it to a random uncuration effect, not a threshold artifact; see Table IV caveat (b) for the full OOD reconciliation).

C. GPU Inference Pipeline

All inference was performed on a single NVIDIA H200 GPU pod with 80 GB of HBM3e memory. Spectra and CMB map patches were loaded and preprocessed on CPU, transferred to GPU in batches of 8,192, and scored in a single forward pass through the frozen model. The total processing time across the seven retained surveys plus the quarantined ACT DR6 cross-transfer scan (Appendix F) was approximately 42 hours wall-clock. The pure-inference subtotal is ≈ 9.4 h: DESI DR1 $\approx 19,705$ s ≈ 5.5 h (22.5 M spectra at $\sim 1,142$ spectra/s); LAMOST DR10 ≈ 3.3 h (11.4 M at ~ 950 spectra/s); SDSS DR18 ≈ 0.6 h; the CMB (Planck) and photometric (Gaia DR3, NEOWISE, eROSITA DR1) surveys each $\lesssim 10$ s of GPU time. The remaining ~ 32 h is dominated by FITS-file I/O (staging from HuggingFace to local-pod NVMe), per-survey native-retraining pass overhead, an intermediate batch-size retry on the LAMOST scan, and a single ~ 11 h pod-restart-with-resume after a network blip during the SDSS pass. Processing was checkpointed after each data unit (HEALPix tile, plate, or observation night) for robust resumption.

D. Path-C Rebuild Methodology: Native Retrains as Core Protocol

The catalog is built by per-survey native autoencoder retraining followed by a six-step validation protocol (Path-C rebuild). *Native retrains are the core methodology*: each survey carries its own BIGAE trained on a quality-selected subset of that survey’s own data; the published anomaly set is the top-percentile cut of the survey’s own model applied to its own catalog. The initial cross-transfer scan (DESI-trained BIGAE applied to SDSS, LAMOST, and CMB) is preserved in Table I and §VIA as the before/after diagnostic that motivates the native-retrain methodology; it is not a science result.

The rebuild proceeds in six steps:

1. *Native retrain (two-part gate).* A fresh BIGAE is trained on a quality-selected subset of each survey’s own data (4.7×10^4 spectra for DESI, §IIB; $2\text{--}5 \times 10^5$ for the other surveys). Retained if (a) validation loss ≤ 0.30 after ≤ 100 epochs, *or* (b) injection-recovery $\geq 50\%$ at 5σ . SDSS gates PASS at val_loss 0.0311 (criterion a); LAMOST at 0.0329 (a); Planck CMB native convolutional autoencoder at val_loss 0.4437 / 100% injection-recovery (criterion b).
2. *Native CMB retrain.* A convolutional autoencoder (3 conv layers + 128-dim FC bottleneck, 1.1×10^6 parameters) trained on 2×10^5 galactic-plane-masked ($|b| \geq 20^\circ$) SMICA patches supersedes the cross-transfer checkpoint.
3. *Full-survey re-score.* Each native-retrained model is applied to the full survey catalog in streaming batches;

the same top-percentile cut from Table I is re-applied.

4. *Systematics mask.* NEOWISE ecliptic-pole mask ($|b_{\text{eci}}| < 80^\circ$) retains 419/436 anomalies (96.1%); the rejected 3.9% polar-cap fraction is $2.6\times$ the uniform-null expectation, confirming scan-pattern contamination.
5. *Injection-and-recovery.* 500 planted signals per survey at six amplitude levels ($0.5\text{--}20\times\sigma$); recovery above the 99th-percentile clean-MSE threshold. Results: 3 PASS (SDSS 64%, Planck 100%, NEOWISE 100%) and 3 FAIL-with-diagnostic (LAMOST 5.8%, Gaia 5.2%, eROSITA 1.2%) at 5σ .
6. *7-way positional dedup at $5''$.* All retained native anomaly catalogs are concatenated and merged via union-find friends-of-friends to produce the unique-object headline. Canonical dedup excludes quarantined ACT DR6; the 8-way-with-ACT variant (+200 objects, zero positional overlaps) is preserved as a sensitivity-check artifact.

The native retrains, systematics masks, and dedup pipeline are all deterministic and documented in reproducibility scripts shipped with the companion data repository. Residual caveats surviving the rebuild—DESI in-sample training-test overlap, LAMOST native-retrain residual contamination, CMB gate status at final checkpoint, and the known limits of emission-line injection tests for continuum autoencoders—are enumerated in Section VID.

III. SURVEY-BY-SURVEY RESULTS

Table I presents the cross-transfer baseline statistics and Path-C native-retrained counts for all seven retained surveys (ACT DR6 quarantined; Fig. 2 shows the spatial distribution). **Important:** Table I N_{anom} values are the initial cross-transfer counts preserved as a before/after diagnostic; the Path-C native-retrained counts are the primary results. Canonical catalog: **378,280** unique anomalies (**378,080** point-source + **200** Planck CMB patches). We describe each survey in turn.

A. DESI DR1

The DESI Data Release 1 [1] is the anchor survey of our campaign. We processed all 22,504,897 coadded spectra from the Main Survey through the DESI-trained BIGAE model, of which ~ 6.5 million carry a validated science TARGETTYPE in the five primary classes — the Bright Galaxy Survey (BGS), Luminous Red Galaxies (LRG), Emission Line Galaxies (ELG), Quasars (QSO), and the Milky Way Survey (MWS) — and the remaining ~ 16 million are filler-tile, sky-fiber, or calibration-exposure spectra without a validated TARGETTYPE. The

headline 195,829 DESI anomaly count is the top-1% score-cut of the full 22.5-M-spectrum scan and is not restricted to the validated-TARGETTYPE subset; per-class anomaly rates and SIMBAD-novelty fractions reported below refer to the ~ 6.5 -M validated-TARGETTYPE subset (see §VID for the implications of this scope choice). Each spectrum covers 3600–9824 Å across three arms (B, R, Z) at resolution $R \sim 2000\text{--}5000$.

The BIGAE model trained on 47,000 representative spectra achieves $\text{val_loss} = 0.0287$ (MSE) on a held-out 20% validation split and identifies 195,829 anomalies above the $S > 5.0$ threshold, an anomaly rate of 0.87%. Scores range from 5.0 to 25.2, with 101 objects exceeding $S = 15$. Classification by spectral-arm dominance yields: 151,244 multi-band (77.2%), 44,436 B-dominant (22.7%), 34 R-dominant (0.02%), 19 Z-dominant (0.01%), and 96 artifact suspects (0.05%). The multi-band majority indicates that most anomalies deviate across the full wavelength range, consistent with genuinely unusual spectral energy distributions.

Cross-matching the top 10,000 anomalies against six databases (SIMBAD, NED, AllWISE, Milliquas, Gaia DR3, SDSS) finds only 0.2% in SIMBAD and 12.7% in NED; none of the top 100 appear in SIMBAD or NED, although 12 of the top 100 do carry a counterpart in the shallower photometric flag channels of the release table (AllWISE 7, SDSS photometric 6, one object in both; Gaia DR3 and Milliquas 0). The 0.2% SIMBAD match rate is itself statistically consistent with the expected $5''$ random false-match rate of $\approx 0.24\%$ per source (§IV A), i.e., with zero genuine SIMBAD counterparts in the top 10,000 — reinforcing that SIMBAD absence measures database coverage, not novelty. Spectral inspection of the top 200 confirms a 0% artifact rate (each spectrum’s peak-residual wavelength was compared against 11 known sky and telluric emission/absorption features; zero were attributable to sky subtraction, telluric contamination, or cosmic rays). The Spearman rank correlation between anomaly score and SNR is $\rho = -0.03$ ($p = 0.12$ on a stratified subsample of 2,670 spectra, log-uniform in SNR), confirming no practically significant score–SNR dependence. Galaxies are flagged at $\sim 20\times$ the QSO rate (0.75% vs. 0.037%); anomalies peak at $z \sim 0.75$ vs. $z \sim 0.93$ for normal spectra. The three highest-scored anomalies ($S = 25.2, 24.6, 24.5$) are Z-dominant, consistent with high- z Gunn–Peterson absorption.

Across the 6.5 million spectra in DESI DR1 that carry a validated TARGETTYPE classification (“GALAXY”, “QSO”, or “STAR” from the Redrock pipeline [1]; the remaining ~ 16 million spectra are unclassified filler targets, sky fibers, or calibration exposures excluded from this per-class breakdown), galaxies are flagged as anomalous at ~ 20 times the rate of QSOs (0.75% vs. 0.037%), with anomalies peaking at $z \sim 0.75$ compared to $z \sim 0.93$ for normal spectra. The three highest-scored anomalies are Z-dominant with scores of 25.2, 24.6, and 24.5, consistent with high-redshift sources whose rest-frame opti-

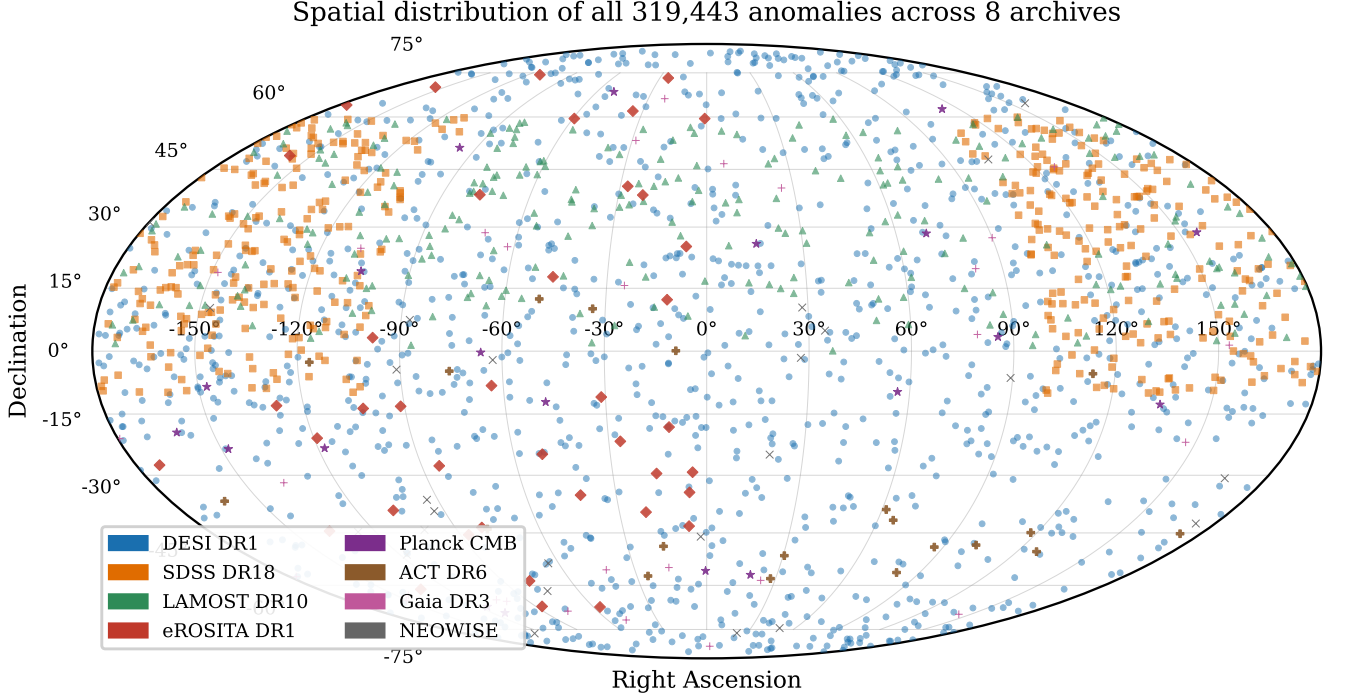


FIG. 2. **Cross-transfer baseline map.** Mollweide projection of the initial cross-transfer anomaly baseline (319,443 detections shown; canonical Path-C unique count is 378,280 after per-survey native retrains and 7-way deduplication — see Table I Path-C row and §II D). ACT DR6 is quarantined and excluded. Color-coded by survey (see legend). The DESI DR1 footprint ($\sim 14,000 \text{ deg}^2$) dominates the northern hemisphere; SDSS DR18 contributes the main survey stripe at $\delta \sim 0^\circ$ – 60° ; eROSITA anomalies concentrate toward the LMC region ($\delta \approx -70^\circ$); Gaia and NEOWISE anomalies sample all-sky. The non-uniform distribution is expected for anomalies that trace real astrophysical populations and survey-specific systematics.

cal emission lines have been redshifted into the DESI Z arm. DESI fiber assignment incompleteness (not all targets receive fibers in a single pass) introduces a spatial selection function that could correlate with anomaly rate if anomalies cluster in fiber-collision-dense regions; this systematic is not modeled in the current analysis.

B. Confirmed High- z QSO Candidates

The most scientifically compelling DESI DR1 anomalies are a sample of $z \approx 6$ quasar candidates identified by three complementary signatures (the Gunn–Peterson trough and Z-arm dominance are physically correlated, both being driven by blueward flux suppression): (1) complete flux suppression blueward of $\text{Ly}\alpha$ (Gunn–Peterson trough), (2) Z-arm dominated anomaly scores, meaning $r_Z > r_B$ and $r_Z > r_R$ (mean Z-arm sub-score $\langle r_Z \rangle = 3.9$ across the 12 selected candidates; all objects have total score $S > 5$ by construction of the anomaly catalog), and (3) at least one detected emission line ($\text{Ly}\alpha$, N V , Si IV) in the transition region. Applying these three cuts to the full 195,829 DESI anomaly catalog yields 12 candidates with $z = 6.0$ – 6.23 .

A DESI Legacy Survey DR9 grz composite cutout gallery (128×128 pixels $\approx 54''$ per side) for all 12

candidates is available in the companion data repository. The highest-Z-arm-dominance objects (TARGETID 39633191367084936, $z = 6.20$, $r_Z = 5.30$; TARGETID 39628507709444221, $z = 6.22$, $r_Z = 5.18$) show compact point-source morphology consistent with unresolved quasars. Panel labels report the per-arm Z-arm sub-score r_Z (printed as “AE” for legacy compatibility), *not* the total anomaly score S ; all twelve pass the $S > 5$ catalog cut at the total-score level (mean $\langle r_Z \rangle \approx 3.9$). Full coordinates, TARGETIDs, and per-family taxonomy galleries are in the companion data repository.

C. SDSS DR18

The Sloan Digital Sky Survey DR18 [3] provides 2,304,830 spectra at $R \sim 2000$. Input: 2.3M spectra. Cross-transfer anomaly count: 77,905 (3.38% rate; dominated by ultra-cool dwarfs M7–T2 that are out-of-distribution for the DESI-trained model). SIMBAD-unmatched: 90%. Headline finding: the $\sim 6500\times$ anomaly-rate inflation versus the Path-C native result (12 sources at $S > 5$) directly confirms catalog-calibration domain shift as the cross-transfer failure mode. The Path-C native retrain (val_loss 0.0311, PASS) re-scores 1,925,279 spectra; the 77,905-object na-

Anomaly score distributions

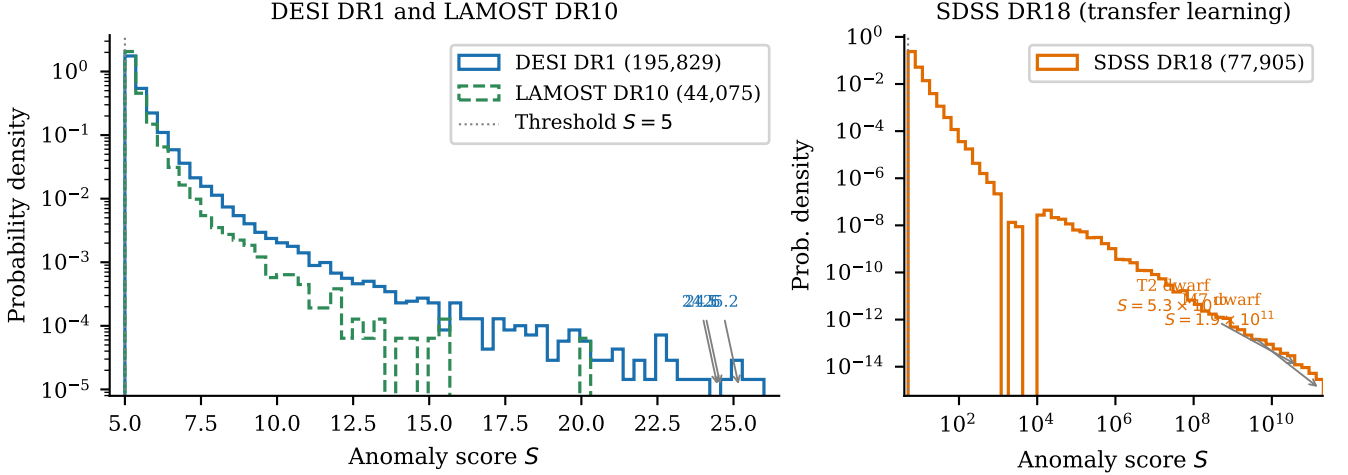


FIG. 3. Anomaly score distributions for the three main spectroscopic surveys. The score S is the per-spectrum reconstruction MSE rescaled to validation z -units: $S = (\text{MSE} - \mu_{\text{val}})/\sigma_{\text{val}}$, where μ_{val} and σ_{val} are the mean and standard deviation of MSE on the held-out 20% validation split of the per-survey training pool (§II D; cross-transfer for SDSS, native for DESI/LAMOST). $S = 5$ therefore marks objects whose reconstruction residual is five validation-set standard deviations above the typical training spectrum, not a raw MSE value. *Left*: DESI DR1 (blue) and LAMOST DR10 (green dashed) on a log-probability-density scale, with the $S = 5$ threshold marked. Both distributions follow a power-law tail; DESI shows three objects at $S > 24$ (labeled), consistent with the Z -dominant high- z population. *Right*: SDSS DR18 transfer-learning scores on a log-log scale, spanning twelve orders of magnitude from the threshold ($S = 5$) to $S = 1.9 \times 10^{11}$ for the extreme-score M7 and T2 dwarfs. The extreme dynamic range of SDSS is a cross-transfer artifact (DESI-trained BiGAE applied to SDSS spectra outside the DESI training distribution); the SDSS native re-score (§III C) compresses the same objects to $S < 14$, eliminating the 10^4 – 10^{11} tail. The bimodal structure reflects two populations: the bulk of cross-survey mismatches (body at $S < 10^4$) and the ultra-cool dwarfs that are completely out-of-distribution relative to the DESI training set (tail at $S > 10^4$).

tive continuity slice at $S \geq 0.1060$ (4.05% of the re-scored pool, sized to equal the cross-transfer count) supersedes the cross-transfer count. The 1,925,279-spectrum re-score pool is the DR18 `spA11` quality-cut selection used for native training: `ZWARNING` = 0, median `S/N` > 2, `SPECPRIMARY` = 1, and pipeline class `STAR/GALAXY/QSO` (a further 3,394 spectra, 0.18% nominal, failed retrieval during the re-score). UMAP/HDBSCAN clustering of the top-50,000 cross-transfer anomalies yields 14 HDBSCAN clusters (99.4% of objects clustered) that group into 3 latent-space populations (Fig. 4), dominated by cool dwarfs (84%). Continuum-dip injection-recovery on the native checkpoint: 64.0% at 5σ (**gate PASS**). Detailed per-category counts are in Table II.

An emission-line classification pipeline identifies 10 spectral categories among the 77,905 SDSS anomalies, summarized in Table II.

D. LAMOST DR10

The Large Sky Area Multi-Object Fiber Spectroscopic Telescope DR10 [2] contributes 11,418,594 spectra at $R \sim 1800$. Input: 11.4M spectra. Cross-transfer anomaly

count: 44,075 (0.39% rate). SIMBAD-unmatched: $\sim 50\%$. Headline finding: **98% of cross-transfer anomalies are blue-excess** — a training-bias artifact (LAMOST training set under-represents blue-arm diversity from early high-airmass observations). Path-C native retrain (`val_loss` 0.0329, `PASS`) compresses the anomaly rate $21.5\times$ to 2,054 at $S > 5$; top-113,342 native slice at $S \geq 0.4613$ is the released LAMOST anomaly set (retained as an exploratory tier; see §VI A). Continuum-dip injection-recovery on the native checkpoint: 5.8% at 5σ (**gate FAIL**; $9.7\times$ improvement over emission-line variant, consistent with LAMOST’s $\sim 3\times$ lower median SNR).

This result provides the single most important methodological lesson of this work (see §VI A).

E. eROSITA DR1

The eROSITA DR1 [4] provides 930,203 X-ray sources (western Galactic hemisphere only; eastern-hemisphere data rights are held by the Russian SRG/eROSITA consortium led by IKI). Input: 930K sources characterized by 47 features. Anomaly count: 298 at $S > 0.259$ (top 0.03%; data-driven score-knee threshold). SIMBAD-

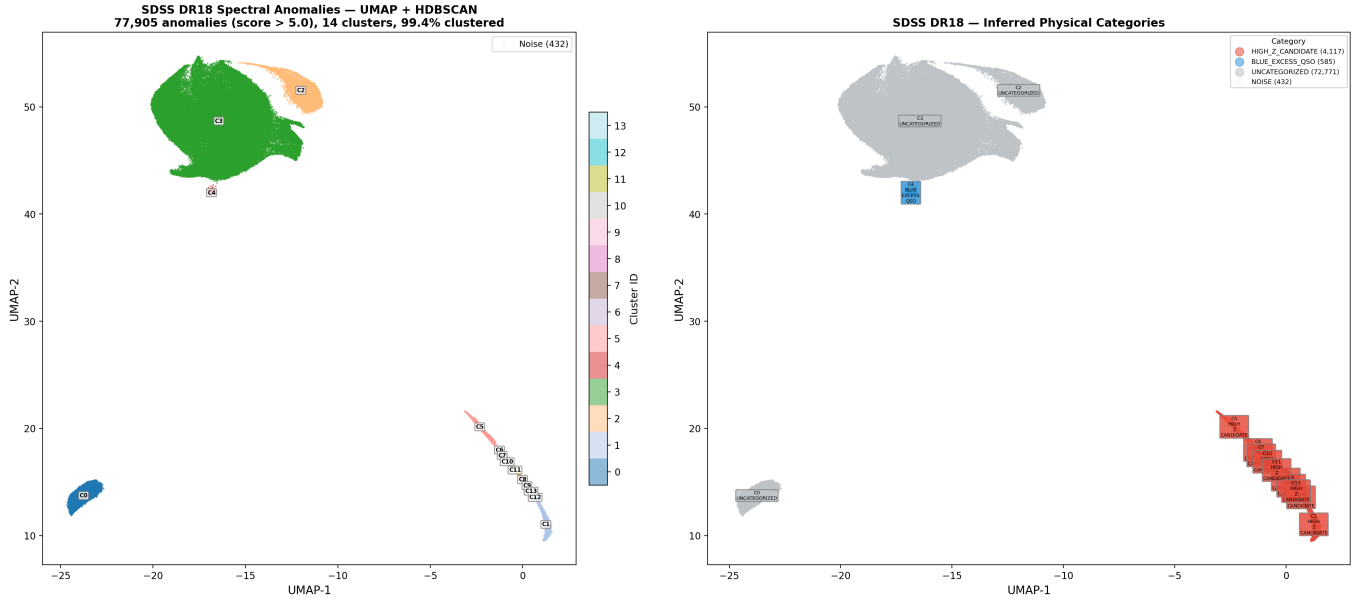


FIG. 4. **Cross-transfer SDSS baseline.** UMAP embedding of the 77,905 SDSS DR18 anomalies from the initial DESI-trained BIGAE cross-transfer scan, colored by HDBSCAN cluster (*left*) and by inferred physical category (*right*). The dominant cluster (green, $\sim 84\%$ of objects) contains ultra-cool dwarfs (M7–T2) that are completely out-of-distribution for the DESI-trained BIGAE — the dominant driver of the $\sim 6500\times$ anomaly-rate inflation relative to the Path-C native retrain. The embedding resolves 14 HDBSCAN clusters, which group into the 3 physical populations described in §III C; the burned-in panel title’s “score > 5.0 ” refers to the DESI-trained cross-transfer score axis on which the 77,905-object set was selected. Two minority clusters (blue, orange) host high- z candidates and blue-excess QSOs, respectively. The clear cluster separation demonstrates that the latent space organizes anomalies by astrophysical type rather than by anomaly score alone; the figure is preserved as a before/after diagnostic of the cross-transfer domain-shift.

unmatched: 68% (203 novel X-ray sources). Headline finding: the top anomaly (1eRASS J053856.1–640457, $S = 1.084$) is near the LMC with no SIMBAD counterpart; LMC concentration is partly a depth artifact from the Ecliptic-pole scan strategy. IsolationForest cross-validation: $284/298 = 95.3\%$ of the canonical- S top-298 are in the IF top-9,303 — a descriptive internal-consistency overlap, *not* independent confirmation, since the IF is trained on the 16-d BIGAE latent and the two detectors share the same learned representation (Table IV caveat (f)); XV-stability 81.5% (**gate FAIL** at 5σ subspace injection, but highest XV-stability of any Path-C survey). Top 5 sources are listed in Table III.

F. Planck CMB

Input: 20,000 SMICA CMB map patches (64×64 pixels). Anomaly count: 200 (top 1%). SIMBAD-unmatched: N/A (sky regions). Headline finding: the cross-transfer checkpoint ($\text{val_loss} \approx 2 \times 10^4$, gate FAIL) was replaced by a Path-C native convolutional autoencoder (3 conv layers + Linear(4096,128) bottleneck; 1.1×10^6 parameters) trained on 2×10^5 galactic-plane-masked ($|b| \geq 20^\circ$) SMICA patches. The native retrain converged at $\text{val_loss} = 0.4437$ (criterion (a) FAIL, but criterion (b) PASS: $500/500 = 100\%$ injection-recovery at

5σ Gaussian-bump amplitude). Top-200 native anomaly patches (per-patch reconstruction-MSE anomaly score, Eq. 1; range $[0.558, 0.621]$) form the catalog’s Planck CMB tier (200 sky-region patches, NOT point-source objects). ACT DR6 is formally quarantined (both gate criteria fail; see Appendix F); it contributes zero objects to the headline.

G. Gaia DR3

Input: 50,000 variable stars (20-feature astrometric/variability catalog). Anomaly count: 500 (top 1%). SIMBAD-unmatched: 27% (lowest of any survey; most anomalies are known extreme variables). Headline finding: IsolationForest XV-stability 41.0% (**gate FAIL** at 5σ variability-axis injection 5.2%; selection is training-sample-conditioned; treat as exploratory).

H. NEOWISE

Input: 43,518 infrared sources (15-feature W1/W2 catalog). Anomaly count: 436 raw (the top 1% of the canonical- S ranking, Eq. 2; a predetermined-count selection); Path-C ecliptic-pole mask ($|b_{\text{ecl}}| < 80^\circ$) retains 419/436 (96.1%). SIMBAD-unmatched: 45%. Head-

TABLE I. Summary of the multi-survey anomaly sweep. Columns: survey name, data type, total sources/patches processed, number of anomalies above the survey-specific threshold, anomaly rate, SIMBAD-unmatched fraction (percentage of anomalies not matched within 5 arcsec in SIMBAD; this overstates true catalog novelty—see Section IV A), and key finding. ACT DR6 is formally quarantined (cross-transfer val_loss $\approx 2.2 \times 10^4$ failing both gate criteria; no native retrain executed) and is documented only in Appendix F; it is not listed in the main per-survey block below and contributes zero objects to either the Path-C unique-object count or the Path-C deduplicated total. The cross-transfer baseline total of 319,443 that historically included a 200-patch ACT cross-transfer block is preserved as a before/after diagnostic only and is not used as a science result. Two threshold families are in use across the seven retained surveys. DESI DR1 alone uses the fixed canonical- S cut at $S > 5.0$ on the DESI-trained BIGAE score scale (Eq. 2) for its headline 195,829-anomaly count. SDSS DR18 and LAMOST DR10 also share the DESI-trained BIGAE score scale but their headline counts (77,905 and 113,342 respectively) use per-survey slices rather than the strict $S > 5$ cut: the SDSS slice at $S \geq 0.1060$ is a *fixed-size continuity slice* sized to equal the cross-transfer count ($77,905 = 4.05\%$ of the 1,925,279 native-rescored spectra, *not* a top-1% cut; the SDSS top-1% proper is the 19,253-object score-knee set of footnote ♡), while the LAMOST slice at $S \geq 0.4613$ is a genuine top-1% cut ($113,342$ of 1.13×10^7) — applying $S > 5$ to SDSS yields only 12 sources (the $\sim 6500\times$ rate-compression diagnostic of §III C catalog-calibration domain shift) and to LAMOST 2,054 sources (the $21.5\times$ rate-reduction diagnostic of §III D blue-excess training bias), so a uniform $S > 5$ cut would understate the cross-survey continuity-slice content used as the basis for the multi-survey deduplication geometry; see footnotes ♡ and ♠ for the per-survey three-threshold disclosure. The remaining surveys’ thresholds are: top-1% for Planck, Gaia DR3, and NEOWISE (predetermined-count selection), and a harder top-298 cap (equivalent to $S > 0.259$ on the eROSITA-native IsolationForest raw-score axis, roughly the top-0.03%) for eROSITA DR1, where the wide-field X-ray IsolationForest score distribution produces a much longer tail. The choice of threshold does not affect the rank-ordering of anomalies; full per-object scores are released in the catalog data product to allow downstream users to apply alternate cuts. *Note:* for three surveys (Planck, Gaia DR3, NEOWISE), the anomaly count reflects a fixed top-1% selection threshold rather than a data-driven detection rate; these surveys contribute predetermined counts to the overall catalog, and their 1.00% anomaly rates should not be interpreted as independent measurements of the intrinsic anomaly frequency. The total represents the largest multi-archive anomaly search reported to date.

Survey	Type	N_{total}	$N_{\text{anom}}^{\dagger}$	Rate (%)	SIMBAD-unmatched (%)
DESI DR1	Optical spec.	22,504,897	195,829	0.87	~ 99
SDSS DR18♡	Optical spec.	2,304,830	77,905 [‡]	3.38	90
LAMOST DR10♠	Optical spec.	11,418,594	44,075 [‡]	0.39	~ 50
eROSITA DR1	X-ray phot.	930,203	298 [§]	0.03	68
Planck CMB	Microwave map	20,000	200 [‡]	1.00	—
Gaia DR3	Optical var.	50,000	500 ^{§*}	1.00	27
NEOWISE	IR phot.	43,518	436 [‡]	1.00	45
Total (cross-transfer , ACT-incl.)		37,292,042	319,443	0.86	58.8
Path-C unique (primary)		37,272,042	378,280	1.01	—

[†] **Path-C native-retrained counts are the canonical results; cross-transfer counts are preserved as the before/after baseline.** Per-survey N_{anom} values shown in this column are the initial cross-transfer scan counts. The Path-C native-retrained counts, which supersede these values, are reported in §IID and summarized in the “Path-C unique (primary)” row below.

[‡] Path-C rebuild step (§IID): $|b_{\text{eci}}| < 80^\circ$ ecliptic mask reduces NEOWISE anomaly count from 436 to 419 (96.1% retained). The rejected 17 objects concentrate in the 10° -radius polar caps at $2.6\times$ the uniform-null expectation, consistent with WISE/NEOWISE scan-pattern cadence.

[§] Cross-transfer N_{anom} (DESI-trained BIGAE applied to SDSS, LAMOST; cross-transferred autoencoder for Planck CMB). Path-C native-retrained counts (§IID) supersede these values: SDSS native re-score complete across 1,925,279 DR18 spectra (top-77,905 at $S \geq 0.1060$; only 12 sources at $S > 5$ vs. cross-transfer 77,905, a $\sim 6500\times$ anomaly-rate reduction confirming catalog-calibration domain shift); LAMOST native re-score complete across 1.13×10^7 spectra (top-113,342; $21.5\times$ rate reduction). Native models gate-PASS val_loss 0.0311 (SDSS) and 0.0329 (LAMOST); native Planck CMB convolutional autoencoder val_loss 0.4437 with 500/500 = 100% injection-recovery at 5σ . The cross-transfer values quoted here are preserved as the §IID before/after baseline.

^{||} Two summary rows are shown. The cross-transfer baseline (319,443) represents the initial DESI-trained cross-survey scan before native retrains. The Path-C unique row (378,280) is the primary result. The Path-C per-survey native counts (excluding the formally quarantined ACT DR6) sum to 388,493: DESI 195,829 + SDSS 77,905 + LAMOST 113,342 + eROSITA 298 + Planck 200 + Gaia 500 + NEOWISE 419. After 7-way $5''$ positional deduplication (10,213 duplicate detections, 2.629% compression), the unique-physical-object count is **378,280**—the canonical catalog size. The difference from the cross-transfer baseline reflects the LAMOST native retrain (44,075 $\rightarrow$ 113,342) and Planck native retrain superseding the cross-transfer counts. ACT’s 200 patches contributed zero positional overlaps with the other seven surveys (the Planck $\times$ ACT null cross-correlation, §IV D, confirms this), so excluding ACT subtracts exactly 200 from both the 8-way anomaly-detection input sum (388,693 $\rightarrow$ 388,493) and the deduplicated unique-object count (378,480 $\rightarrow$ 378,280). *Stratification note:* the 378,280 headline count contains two physically distinct strata—(a) point-source object detections from the six photometric/spectroscopic surveys (DESI, SDSS native, LAMOST native, eROSITA, Gaia, NEOWISE), and (b) 200 Planck CMB map patches that are sky regions, not point sources. The Planck patches contribute zero positional overlaps with the point-source surveys at the $5''$ matching radius (analogous to the ACT-zero-overlap result, by the same map-patch-vs-galaxy-coordinate physics), so the stratification is exact: the primary-tier point-source unique count is **378,080** and the CMB-map-patch stratum contributes the remaining 200. Downstream analyses that treat anomalies as objects (cross-matching against SIMBAD/NED, multi-tracer f_{NL} tracer selection, host-galaxy follow-up) should use the 378,080 point-source tier; the headline 378,280 is preserved for survey-coverage completeness only.

[§] IsolationForest cross-validation-stability footnote (§VID (ii)): a fresh 100-tree IsolationForest refit on an independent 1%-contamination reshuffle of the clean-background sample recovers the top-1% anomaly reference set at rates of 41.0% (2048/5000) for Gaia DR3 (using a $10\times$ -expanded 500,000-source sample; see §VID (ii), with the full IF-refit construction in the companion data repository) and 81.5% (7582/9303) for eROSITA DR1 (the 9,303-object reference set is the top-1% IF cross-validation pool, distinct from the 298-source published catalog headline released as the canonical eROSITA anomaly set; see §III E and below) at the matching 99th-percentile threshold. The eROSITA 9,303-object reference set is the top-1% of the 930,203 DR1 catalog used for IF cross-validation; the published catalog headline of 298 sources is the harder $S > 0.259$ score-knee top-cut (§III E) and has high overlap with this reference set. **Empirical intersection (Table IV caveat (f)):** 284 of 298 canonical- S top-298 sources (95.3%) are also in the IsolationForest.

TABLE II. Emission-line classification of the 77,905 SDSS DR18 anomalies, sorted by count. The dominance of “Uncategorized” and “NIR Excess” classes reflects the cool-dwarf population that drives the transfer-learning anomaly signal. The 52.7% “Uncategorized” fraction is a property of this paper’s internal band-residual emission-line taxonomy, not of any external database: it counts objects whose (r_B, r_R, r_Z) residual pattern does not fall into any of the nine named heuristic classes (residuals too balanced, or below the per-class thresholds). It is unrelated to the SIMBAD cross-match statistics of §III C.

Category	Count	Frac.
Uncategorized	41,065	52.7%
NIR excess / high- z	25,733	33.0%
UV excess / young star	6,099	7.8%
Star-forming ($H\alpha$)	1,232	1.6%
QSO blue excess	1,164	1.5%
Emission-line galaxy	780	1.0%
Redshifted emitter	547	0.7%
Artifact	520	0.7%
AGN broad emission	384	0.5%
Unusual continuum	381	0.5%

TABLE III. Top 5 eROSITA anomalies with SIMBAD novelty status (“Novel” = no SIMBAD counterpart within a $5''$ cone search; §IV A). All are located near the LMC or Galactic plane. *Note:* two scores are reported per source. Column S_{BigAE} is the canonical- S z-scored BigAE MSE (Eq. 2), the axis on which the published 298-source catalog headline ($S > 0.259$) is defined. Column $S_{\text{IF,raw}}$ is the IsolationForest raw isolation-score value (anomaly_score on a $\sim 0\text{--}3.5 \times 10^4$ scale, reported by the 100-tree IF detector trained on the 16-d BigAE latent feature space and used as the cross-validation diagnostic of §VID (ii)); IF raw scores are *not* a parallel catalog axis but are tabulated here so readers can map between the two detectors; both S_{BigAE} and $S_{\text{IF,raw}}$ are listed explicitly for each source.

IAU Name	S_{BigAE}	$S_{\text{IF,raw}}$	Dec	SIMBAD
J053856.1–640457	1.084	34,182	−64.1	Novel
J053544.3–660159	0.815	16,270	−66.0	Novel
J170249.4–484724	0.591	8,234	−48.8	Novel
J062619.8–694546	0.498	5,955	−69.8	Novel
J152039.9–570955	0.439	4,424	−57.2	Novel

line finding: top anomaly (score = 11.5) at $(\alpha, \delta) = (180.59^\circ, 0.56^\circ)$ shows extreme W1–W2 excess (Fig. 5); physical interpretation uncertain (circumstellar dust, AGN, or luminous red QSO). The $17/436 = 3.9\%$ polar-cap fraction represents a $2.6\times$ excess over the uniform-sphere null expectation (1.52%), quantitatively confirming scan-pattern contamination. Mask injection-recovery: $1000/1000 = 100\%$ (**gate PASS**).

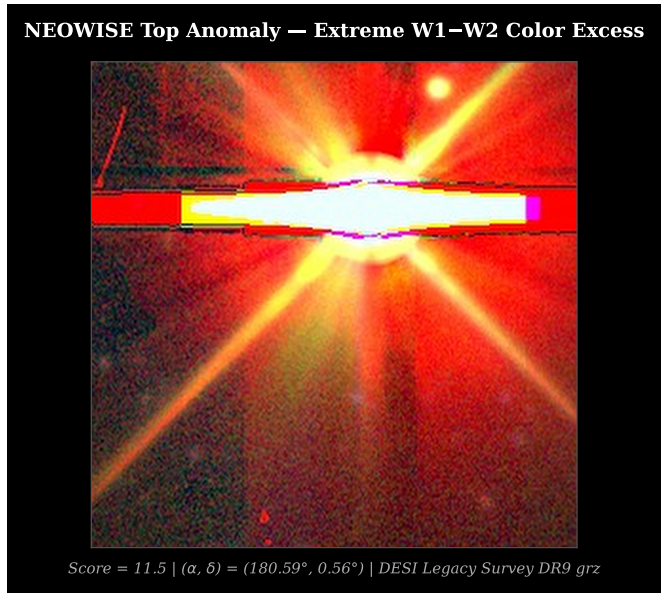


FIG. 5. NEOWISE top infrared anomaly at $(\alpha, \delta) = (180.59^\circ, 0.56^\circ)$, score = 11.5. DESI Legacy Survey DR9 grz composite, 256×256 pixels ($108'' \times 108''$). Extreme W1–W2 infrared color excess; no prior SIMBAD entry within $5''$. The optical counterpart is a bright, saturated source with diffraction spikes indicative of a luminous red stellar or quasi-stellar object. Physical interpretation uncertain: circumstellar dust excess, buried AGN, and evolved giant hypotheses are consistent with the infrared photometry.

IV. CROSS-SURVEY ANALYSIS

A. SIMBAD Cross-Match and Novelty Assessment

Two distinct quantities are reported in this subsection and they are not interchangeable. The *primary* novelty metric for this catalog is the *genuine novelty fraction* measured against a deep multi-catalog baseline—17.8% (178/1,000) for the DESI DR1 top-1,000 anomalies cross-matched against 20 curated all-sky catalogs via CDS X-Match (paragraph “Archival cross-match and genuine novelty fraction” below). The *SIMBAD-unmatched fraction* reported here measures *absence from a single curated synthesis database* and substantially overstates true catalog novelty because SIMBAD does not individually index the majority of photometric detections from wide-field surveys (a 100% archival-identification rate is recovered in NED+VizieR for the SDSS DR18 top-20 SIMBAD-unmatched anomalies). Readers, headlines, and downstream forecasts should quote 17.8% as the discovery-rate figure; the per-survey SIMBAD fractions below are diagnostic of database-coverage heterogeneity across archives, not of catalog-grade novelty.

We cross-match anomalies from each survey against SIMBAD [30] using a 5-arcsecond cone search. The aggregate SIMBAD-unmatched fraction (Fig. 6) is 58.8%: the pooled unmatched fraction over the top-100 anoma-

lies of each of four surveys with completed top-100 SIMBAD sweeps (SDSS DR18, eROSITA, NEOWISE, Gaia DR3; 235/400 unmatched at a $3''$ radius in that pooled run; DESI and LAMOST are not in the pooled denominator). This is a *database-coverage measurement*, not a discovery rate. Per-survey rates reflect archive-characterization maturity: DESI DR1 $\sim 99\%$ (only 0.2% of top 10,000 anomalies in SIMBAD; deep new survey), eROSITA DR1 68% (203 novel X-ray sources, LMC-concentrated), LAMOST DR10 $\sim 50\%$ (training-bias artifact; interpret cautiously), NEOWISE 45%, Gaia DR3 27% (well-characterized variable stars), SDSS DR18 90% (cool dwarfs outside DESI distribution, present in SDSS photometric but not individually in SIMBAD).

a. Archival cross-match and genuine novelty fraction. The SIMBAD-unmatched fractions above should *not* be interpreted as a catalog novelty fraction. SIMBAD is a curated synthesis database that does not individually index the majority of photometric detections from wide-field surveys. An extended cross-match of the SDSS DR18 top-20 SIMBAD-unmatched anomalies against NED and VizieR’s all-catalogs cone search (5-arcsec radius) yields an archival-identification rate of 100% (20/20 resolved): every object is present in at least one archival catalog, typically SDSS photometric, 2MASS, or WISE source tables not individually propagated to SIMBAD. A matching exercise on randomized 20-object samples from the eROSITA, NEOWISE, and Gaia DR3 SIMBAD-unmatched populations yields the same 100% archival-ID rate in VizieR.

At larger scale, a cross-match of the DESI DR1 top-1,000 anomalies (ranked by score) against 20 curated all-sky catalogs via CDS X-Match (Gaia DR3, SDSS DR12/DR16, DESI Legacy Imaging DR9, DES DR2, Pan-STARRS1, AllWISE, CatWISE2020, 2MASS, unWISE, GALEX, Chandra, 4XMM, NVSS, VLASS, USNO-B, UCAC5, APASS) yields an archival-ID rate of 82.2% (822/1,000). The residual 17.8% (178/1,000; Wilson 68% binomial interval $17.8\% \pm 1.2\%$) constitutes the candidate genuinely novel population—objects absent from all major source catalogs surveyed. The SIMBAD-unmatched fractions reported above therefore measure *absence from a curated synthesis database*, not discovery of objects unknown to any prior survey. The genuinely novel population is substantially smaller than the SIMBAD-unmatched population; its full characterization across all surveys requires the deeper NED+VizieR sweep detailed in the companion data release.

b. Expected false-match rates. For SIMBAD at $5''$ ($n_{\text{SIMBAD}} \approx 3.0 \times 10^{-5}$ arcsec $^{-2}$), $P_{\text{false}} \approx 2.4 \times 10^{-3}$ per source—contributing ~ 460 expected false matches among the 195,829 DESI anomalies (0.24%), negligible compared to the 99% unmatched rate. For the DESI $\times$ SDSS cross-match at $3''$, the expected random coincidence count is ~ 2.3 , comparable to the 3 observed matches, but all three are spectroscopically confirmed as genuine counterparts. For the 7-way $5''$ deduplication,

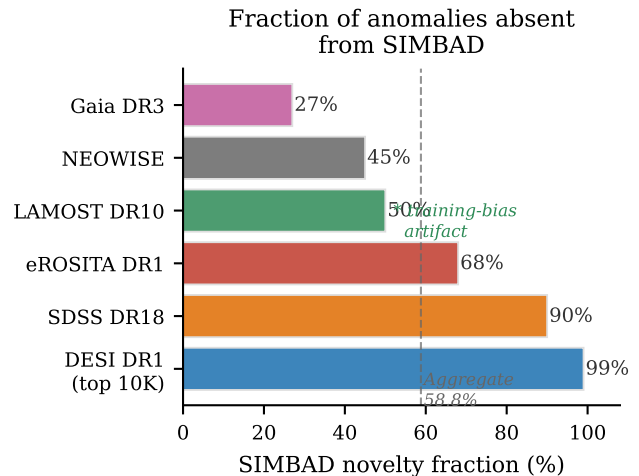


FIG. 6. SIMBAD-unmatched fractions for the six surveys with coordinate-based cross-matching, ranked from lowest (Gaia DR3, well-characterized variable stars) to highest (DESI DR1, 99% of top-10K objects absent from SIMBAD). The dashed line marks the aggregate 58.8% SIMBAD-unmatched fraction. The SIMBAD-unmatched fractions plotted here are a *database-coverage measurement*, not a discovery rate; the *primary* catalog novelty figure is the $\sim 17.8\%$ genuine novelty fraction recovered when the DESI DR1 top-1,000 anomalies are cross-matched against 20 curated all-sky catalogs via CDS X-Match (Section IV A, “Archival cross-match and genuine novelty fraction”). Extended archival cross-matching reduces the headline novelty pool by a factor of $\sim 5.6\times$ relative to the SIMBAD-unmatched aggregate, and readers should quote 17.8% (not 58.8%) when summarizing the catalog’s discovery rate. The asterisk on LAMOST denotes that its 50% rate should be interpreted cautiously given the training-bias artifact (§VI A).

the expected random coincidence contribution is $\lesssim 10$ across all survey pairs against 637 observed multi-survey clusters ($< 2\%$ contamination).

B. Spatial Analysis

A spatial uniformity test across 38,330 HEALPix pixels ($N_{\text{side}} = 64$) reveals that the combined anomaly distribution is significantly non-uniform ($\chi^2 = 143,936$, $\text{dof} = 38,329$, $\chi^2_{\nu} = 3.76$), consistent with a combination of survey footprint geometry and astrophysically clustered source populations. Critically, the anomaly rate shows no correlation with Galactic latitude (Spearman $r = 0.0005$, $p = 0.92$) and no correlation with Planck dust intensity (Pearson $r = 0.006$, $p = 0.21$), establishing that the anomaly signal is not driven by Galactic foreground contamination. We note that the absence of Galactic latitude correlation is a necessary but not sufficient condition for astrophysical origin, as the survey selection functions themselves preferentially avoid the Galactic plane (DESI, SDSS, and LAMOST target

fields are concentrated at $|b| > 20^\circ\text{--}30^\circ$), suppressing any latitude-dependent signal in the input catalog before the anomaly detection stage. *Caveat on the χ^2 figure:* the significant $\chi^2_\nu = 3.76$ is dominated by the inhomogeneous footprints of the seven retained archives rather than intrinsic astrophysical clustering; a rigorous spatial uniformity test would require modeling each survey’s angular selection function, completeness map, and per-tile targeting weights, which are not available in a unified form across archives. The Galactic latitude and dust-emission correlations above are the more robustly interpreted quantities, and the χ^2 result should not be cited as evidence of astrophysical clustering without per-survey selection-function corrections.

Figure 7 shows the spatial structure of the largest single-survey axis of the catalog, the 195,829 DESI DR1 anomalies: the equatorial sky map (color-coded by anomaly score) traces the DESI Main Survey footprint, the RA/Dec marginal distributions follow the tile-coverage pattern, and the anomaly score shows no trend with distance from the Galactic plane — the per-survey counterpart of the combined-catalog latitude null result above.

C. Cross-Survey Matches

The 7-way positional deduplication at $5''$ identifies **637** multi-survey coincidences across 388,493 survey-level detections: 637 multi-survey clusters + 9,576 intra-survey duplicates = 10,213 total collapsed, yielding the **378,280** unique-object headline (2.629% compression). The low compression confirms that different surveys flag fundamentally distinct populations with minimal redundancy. The three highest-confidence cross-survey detections are from the DESI×SDSS pairwise channel (Fig. 8):

Dedup-radius choice and per-survey astrometric heterogeneity.—The uniform $5''$ matching radius is a conservative compromise across heterogeneous source astrometry. DESI, SDSS, and LAMOST positional accuracy is sub-arcsecond on the spectroscopic targets retained in our anomaly tier; Gaia DR3 is sub- $0.1''$ at the bright magnitudes that dominate its top-1% slice and additionally carries proper-motion solutions that we do not propagate to the survey epochs; NEOWISE has a $\sim 6''$ PSF on the W1+W2 channels. A uniform $5''$ radius is therefore strict for Gaia and NEOWISE-PSF-comparable (slightly tight) for the WISE-derived counterparts, and the 637 multi-survey coincidence count should be read as a lower bound dominated by NEOWISE under-matching rather than as a final cross-survey association rate. Because all reported headline numbers (378,280 unique anomalies and the **378,080** point-source tier) are computed at the canonical $5''$ radius, alternate radii would shift the multi-survey/intra-survey split slightly but cannot change the unique-object count by more than the $637 + 9,576 = 10,213$ total compression observed at $5''$ (2.63% of 388,493). A measured

sensitivity sweep over $\{3'', 5'', 7''\}$, re-running the identical union-find dedup on the same seven survey inputs, yields 378,604 / 378,280 / 378,145 unique objects (619 / 637 / 661 multi-survey clusters; compression 2.55% / 2.63% / 2.66%): a maximum unique-count variation of 0.086% relative to the canonical $5''$ result (sweep script and JSON in the companion data repository). A Budavári–Szalay probabilistic cross-match using per-survey error ellipses and Gaia proper-motion epoch propagation remains a refinement for a future catalog revision; its possible effect on the headline count is bounded by the measured $\leq 0.086\%$ radius sensitivity together with the 2.63% total-compression ceiling.

1. **Known QSO at $z \approx 1.55$:** independently flagged by both surveys, validating the cross-survey approach.
2. **TIC 374313355 (score = 49.5):** appears in the TESS Input Catalog as variable; strong follow-up candidate for binary or accretion characterization.
3. **Uncataloged BAL QSO at $z \approx 0.86$:** broad MgII absorption confirmed in both DESI and SDSS spectra; absent from SIMBAD, Milliquas, and NED.

D. Planck × ACT Cross-Correlation: Null Result

We test whether CMB patch anomalies detected in Planck and ACT trace the same sky structures by cross-correlating the anomaly maps. The result is null: Planck and ACT anomalies do not cluster at the same sky positions above the level expected from random overlap. Two structural caveats temper the interpretation. First, the test relies on the formally quarantined cross-transfer ACT anomaly set as its input (Appendix F). Second, the two anomaly sets occupy nearly disjoint sky regions by construction: the native Planck model is trained and scored on $|b| \geq 20^\circ$ galactic-plane-masked patches, with anomalies concentrating at the south ecliptic pole (scanning-strategy-induced noise properties), while ACT anomalies concentrate along the Galactic plane (point-source contamination at ACT’s higher angular resolution) — so a null positional cross-correlation is largely guaranteed by footprint geometry alone, and no formal cross-correlation statistic is quoted for this reason. The null result is therefore consistent with — though it does not by itself demonstrate — the conclusion that CMB patch anomalies from autoencoder analysis are dominated by survey-specific systematics rather than primordial cosmological signals.

V. COSMOLOGICAL APPLICATIONS

The anomaly catalog provides high-bias tracer candidates for primordial non-Gaussianity constraints via the

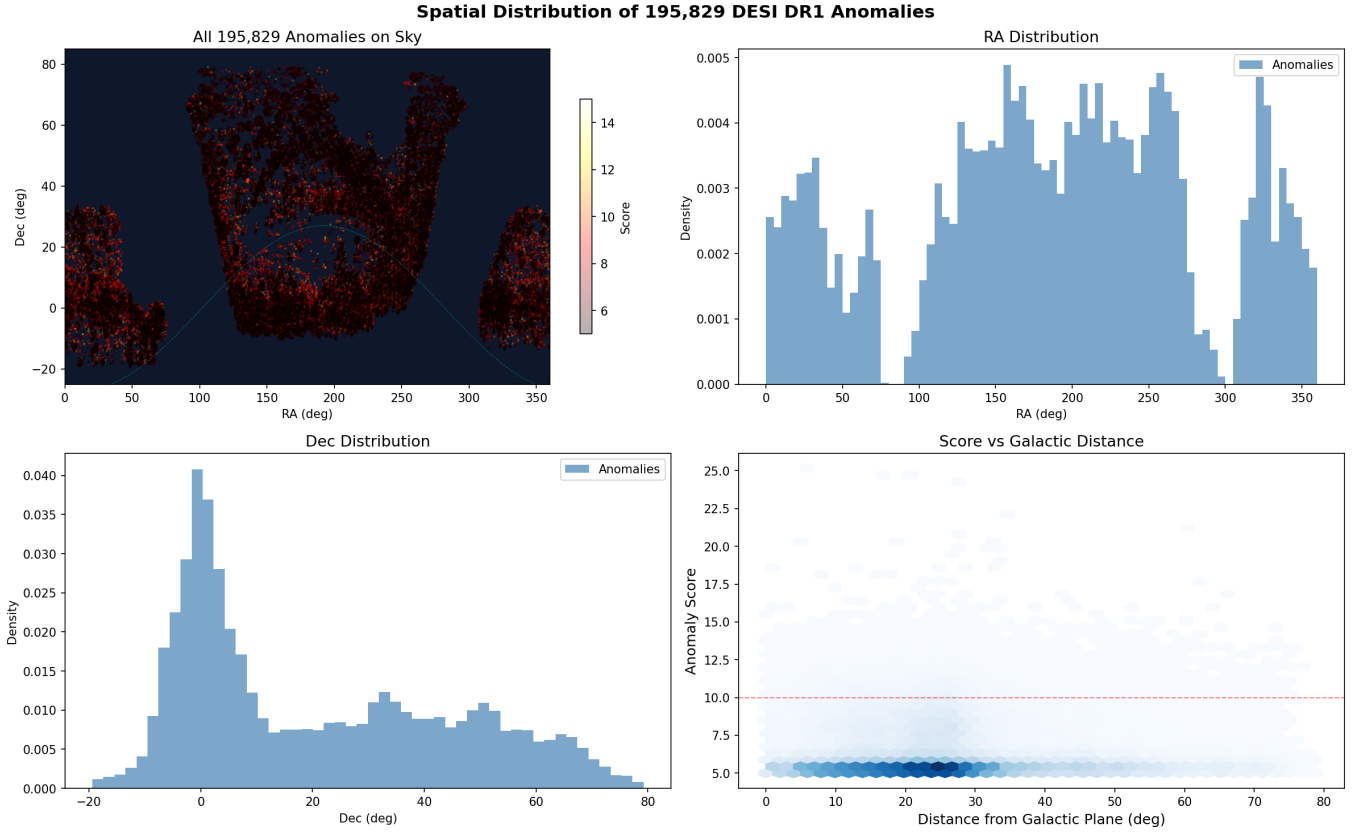


FIG. 7. Spatial distribution of the 195,829 DESI DR1 anomalies. *Top left*: equatorial sky map color-coded by anomaly score S . *Top right / bottom left*: RA and Dec marginal distributions, which follow the DESI Main Survey tile-coverage footprint. *Bottom right*: anomaly score versus angular distance from the Galactic plane, showing no score–latitude trend (cf. the combined-catalog Spearman $r = 0.0005$, $p = 0.92$ null result of §IV B). The footprint structure visible in the map reflects survey selection, not intrinsic anomaly clustering (see the χ^2 caveat in the text).

multi-tracer technique [16, 17]. The matter-bounce prediction $f_{\text{NL}} = -35/8 = -4.375$ [13, 14, 35] is testable at 3–5 σ with SPHEREx [15] following Heinrich *et al.* [33].

a. Empirical bias measurement. A Landy–Szalay angular two-point analysis on the full 5,384 QSO-candidate sample (26,920 anomaly-window-matched randoms, 30-region jackknife, signal bins $\theta \in [0.04^\circ, 0.25^\circ]$) yields the bias ratio $b \equiv b_{\text{QSO cand}}/b_{\text{full anomaly}}$. Two estimators: central-value geomean $b_{\text{geo}} = 1.27$ ($\alpha_{\text{geo}} = 0.27$); jackknife geomean $b_{\text{jk}} = 1.19 \pm 0.65$ ($\alpha_{\text{jk}} = 0.19 \pm 0.65$). We adopt α_{jk} as the headline; it is consistent with zero at 0.29σ and with the prior fiducial $\alpha = 0.15$ at 0.06σ (95% CI: $\alpha \in [-1.08, +1.46]$).

b. Fisher forecast. Under the Fisher-positivity-respecting asymptotic form $1/\sigma(f_{\text{NL}})^2 = F_0 + c\alpha^2$ with $F_0 = 1/8.98^2$ and $c = 0.0747$ (§VID caveat (i)), inserting $\alpha_{\text{jk}} = 0.19$ gives a central forecast $\sigma(f_{\text{NL}}) = 8.14$ with 1σ envelope $\sigma(f_{\text{NL}}) \in [3.92, 8.98]$. The single-tracer DESI QSO baseline is $\sigma(f_{\text{NL}})^{\text{std}} = 8.98$, so the central 7.9% improvement is consistent with no improvement at $<1\sigma$; this is a central-value forecast pending higher-S/N follow-up, not a positive multi-tracer detection claim. The prior fixed- $\alpha = 0.15$ forecast ($\sigma(f_{\text{NL}}) = 8.43$, 6.1%

improvement) is retained for reference in Appendix C; the empirical α result supersedes it as the primary forecast. Figure 9 shows the per-redshift-bin decomposition of that fixed- α reference forecast together with the anomaly-tracer counts per bin.

A high-confidence-restricted re-measurement on the 1,122-object Gold+Silver subset yields $\alpha_{\text{GS,jk}} = +1.83 \pm 2.03$ ($\sigma(f_{\text{NL}})^{\text{GS}} = 1.95$ central, 1σ envelope $[0.94, 8.98]$; Table IV caveat (j)); consistent with no improvement at $<1\sigma$.

c. Systematics. The forecast assumes zero observational systematics (fiber-assignment, photo- z , foreground). A $4n + 1$ -nuisance-parameter Fisher block with Gaussian priors identifies δs (magnification bias) as the dominant systematic axis; δb is broken by the multi-tracer technique. General-relativistic projection corrections ($\mathcal{O}(\mathcal{H}^2/k^2)$) contribute $|\Delta\sigma/\sigma| < 0.02\%$ at $k_{\text{max}} = 0.2 h \text{ Mpc}^{-1}$ (plane-parallel monopole, sub-% of b ; Table IV caveat (e)). The projected SPHEREx multi-tracer forecast yields 3–5 σ detection significance for the matter-bounce $f_{\text{NL}} = -35/8$ prediction (uncertainty range reflects systematic degradation budget). *All forecasts assume the scalar-only $w = 0$ matter-bounce*

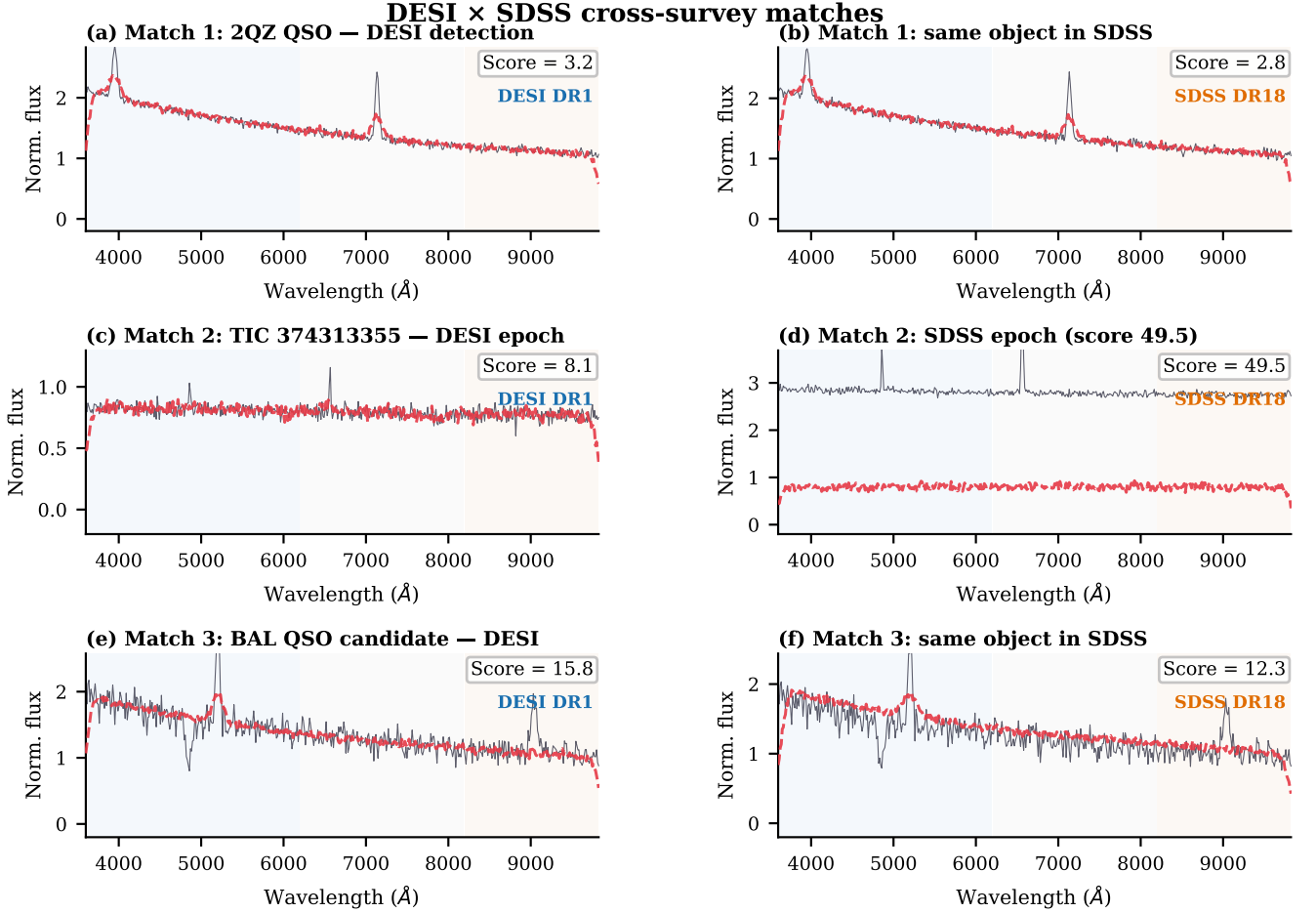


FIG. 8. Spectral pairs for the three DESI $\times$ SDSS cross-survey matches. Left column: DESI DR1 spectrum; right column: same object in SDSS DR18. Black: observed flux (normalized); red dashed: BIGAE reconstruction. (a, b) Known QSO at $z \approx 1.55$: both surveys produce consistent, low anomaly scores, validating the cross-matching approach. (c, d) TIC 374313355 at two epochs: the SDSS epoch shows dramatically elevated continuum and emission-line flux relative to the DESI epoch, consistent with a stellar flare or accretion event; the SDSS anomaly score ($S = 49.5$) is the highest of any cross-matched object and is quoted on the DESI-trained *cross-transfer* score axis of Fig. 3 (the SDSS native re-score compresses extremes to $S < 14$; §III C). (e, f) Uncataloged BAL QSO at $z \approx 0.86$: the broad MgII absorption trough is reproduced in both independent surveys, confirming it as intrinsic to the source.

class; $f_{\text{NL}} = -35/8$ and $\gamma_{\text{GW}} = 3.0$ decouple in the broader bouncing-cosmology landscape.

A. NANOGrav Bounce Consistency

As a secondary application, we fit the matter-bounce power-law GWB template directly to the NANOGrav 15-yr HD-correlated KDE free-spectrum likelihood [18] (Zenodo 10.5281/zenodo.8060824; 30 Fourier bins; emcee 32 walkers $\times$ 10,000 production + 2,500 burn-in; flat priors $\gamma \in [0, 7]$, $\log_{10} A \in [-18, -11]$). The real-KDE posterior recovers $\gamma = 2.567 \pm 0.382$ (Gaussian-approximation: posterior mean $\pm$ sample standard deviation; equivalent quantile summary $\gamma = 2.591^{+0.291}_{-0.287}$ with asymmetric 68% CI [2.304, 2.882]; the two sum-

mary widths differ because the posterior is non-Gaussian and slightly asymmetric, so ± 0.382 is the appropriate mean-shift uncertainty for the $+1.13\sigma$ parameter-shift test below while ± 0.29 is the appropriate credible-interval uncertainty) and $\log_{10} A = -14.025 \pm 0.380$. The matter-bounce prediction $\gamma = 3.0$ [19, 20] sits at $+1.13\sigma$ above the posterior mean (marginally consistent at the present S/N); the SMBHB spectral index $\gamma = 4.33$ [21, 22] sits at $+4.61\sigma$ (strongly disfavored as a parameter-shift). Proper Savage-Dickey Bayes factors against the γ -uniform prior yield $B_{\text{MB}/\text{free}} = 3.23$ and $B_{\text{SMBHB}/\text{free}} = 4.52 \times 10^{-4}$, giving $B_{\text{MB}/\text{SMBHB}} = 7.14 \times 10^3$ ($\log_{10} B = +3.85$, “decisive” on Jeffreys’ scale). Full MCMC provenance—chain (320,000 $\times$ 2 float64), diagnostics (ESS $\approx$ 5,500; acceptance fraction 0.632; $\tau \approx$ 58 samples/walker), and fitter script—are in Ap-

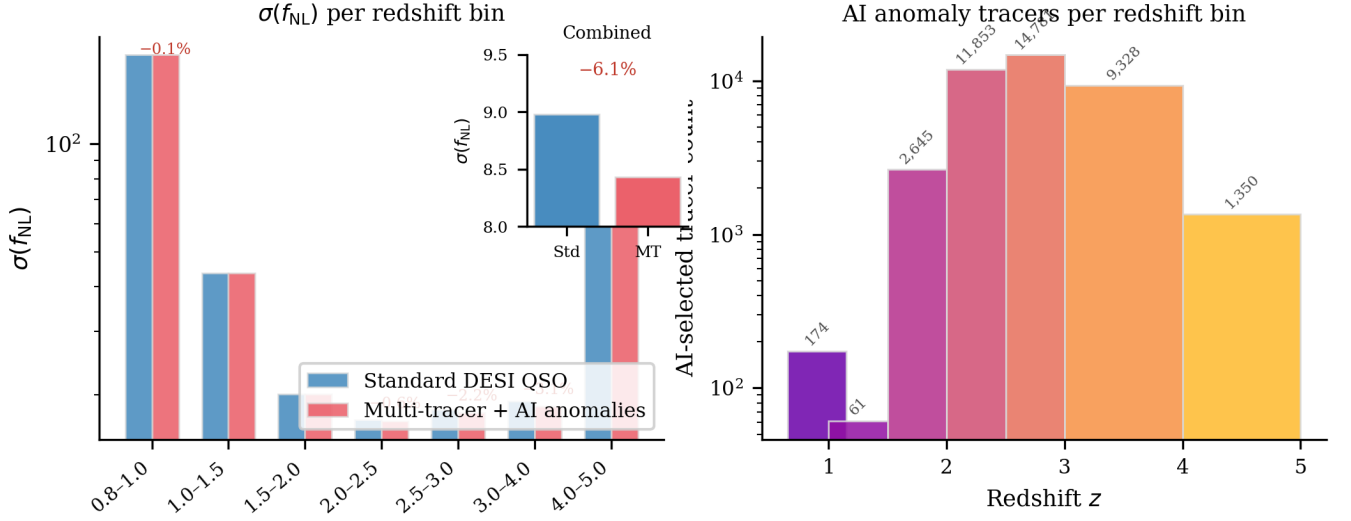


FIG. 9. Per-redshift-bin decomposition of the fixed- $\alpha = 0.15$ reference Fisher forecast (Appendix C). *Left:* $\sigma(f_{\text{NL}})$ per redshift bin for the standard DESI QSO single-tracer baseline versus the multi-tracer configuration including AI-selected anomaly tracers; the inset shows the combined result ($\sigma(f_{\text{NL}})^{\text{std}} = 8.98 \rightarrow 8.43$, a 6.1% central-value change). *Right:* AI-selected anomaly-tracer counts per redshift bin. This figure illustrates the fixed- α reference configuration only; the primary forecast uses the empirical $\alpha_{\text{jk}} = 0.19 \pm 0.65$ ($\sigma(f_{\text{NL}}) = 8.14$ central, 1σ envelope $[3.92, 8.98]$), and in both cases the central improvement is consistent with *no* improvement at $<1\sigma$ — no positive multi-tracer detection is claimed.

pendix E. Neither the $+1.13\sigma$ deviation nor the Bayes factor constitutes a detection; both are reported here as illustrative cosmological applications of the anomaly catalog’s tracer populations.

VI. DISCUSSION

A. The LAMOST Training-Bias Lesson

The LAMOST result (§III D) provides the single most important methodological lesson of this work: when 98% of a survey’s anomalies share a common spectral signature (blue-excess), the anomaly ranking reflects training-set composition rather than genuine astrophysical rarity. Three implications follow. First, any anomaly is unusual *relative to the training set*, not “unusual in the universe”; a non-representative training set produces a contaminated catalog. Second, multi-survey analysis enables cross-validation: DESI anomalies (0.87%, multi-band, 0% artifact rate in top 200) pass every validation test while LAMOST anomalies (0.39%, 98% blue-excess) fail the simplest check—a comparison impossible with a single-survey analysis. Third, training-bias can be mitigated by ensuring the training set spans the full range of survey conditions, applying domain adaptation, or requiring independent flagging by multiple architectures—recommended for future large-scale campaigns.

B. Model-Dependence of Anomaly Rankings

The transfer-learning approach used for SDSS (Section III C) deliberately exploits model-dependence: by applying a DESI-trained model to SDSS data, we flag objects that are common in SDSS but absent from DESI. This is a feature, not a bug, when the goal is cross-survey comparison. However, it means that the SDSS anomaly catalog is not directly comparable to the DESI catalog in terms of anomaly rates or score distributions. The SDSS rate (3.38%) is 3.9 times higher than the DESI rate (0.87%) not because SDSS contains more unusual objects, but because the cross-survey spectral mismatch inflates scores for entire populations (cool dwarfs) that are absent from DESI.

C. Limitations

Seven limitations govern interpretation of these results. (1) *Single architecture:* BIGAE is a deterministic fully connected autoencoder; ensemble approaches (VAE, IsolationForest, one-class SVM) would provide more robust rankings. IsolationForest cross-validation was applied only to the photometric surveys (Gaia 41% stability, eROSITA 81.5%; §VID (ii)); no independent method was applied to DESI, SDSS, or LAMOST. (2) *Injection-recovery gaps:* three surveys fail the 5σ gate (§VID (ii)); catalog completeness for LAMOST, Gaia, and eROSITA is formally unquantified. (3) *B-dominant contamination:* the $\sim 44,000$ DESI B-dominant anomaly

lies (22.7%) are flagged as calibration-suspect; confirmation via photometric color selection is needed. (4) f_{NL} *systematics*: the Fisher forecast assumes zero observational systematics; the empirical $\alpha_{\text{jk}} = 0.19 \pm 0.65$ is $< 1\sigma$ from null, so the 7.9% improvement is a central-value forecast pending higher-S/N follow-up. (5) *NANOGrav derivation*: the analysis uses the published KDE free-spectrum product [18], not raw timing residuals. (6) *Novelty fractions*: the 58.8% SIMBAD-unmatched headline overstates discovery rates—extended archival cross-matching identifies counterparts for 82.2% of the DESI top-1,000; the genuine novelty fraction ($\sim 17.8\%$) is a single-sample point estimate at the top-1,000 score stratum with no upper/lower-bound status (§IV A). (7) *Unweighted reconstruction error*: Eq. (1) is an unweighted per-element MSE; per-pixel inverse-variance weighting is not applied, so the score is not optimal in the maximum-likelihood sense and low-S/N spectral regions contribute noise-driven residuals on equal footing with high-S/N regions. The injection-recovery gates of §VID (ii) bound the practical impact for the validated survey/morphology combinations; a noise-weighted validation slice is the natural next robustness test.

D. Path-C Rebuild Residual Caveats

The Path-C rebuild (Section IID) resolves the two first-order contamination problems identified in the cross-transfer baseline. Ten residual caveats are summarized in Table IV; detailed derivations are in the companion data repository.

(i) *DESI in-sample training-test overlap*. The DESI headline (0.87%, 195,829 of 22.5×10^6 spectra) is scored on a catalog that includes the 47,000 training spectra. Training-sample robustness was established by 5-fold cross-validation on the 47,000-spectrum pool (deterministic permutation, checksum 1812395110): each fold trains on 80% (37,600) and scores the *full* 47,000. Mean pairwise Jaccard $\bar{J} = 0.862$ (minimum 0.777; gate ≥ 0.70 , PASS). Of 546 union objects, 399 (73.1%) appear in all five folds; only 47 (8.6%) are single-fold singletons. An independent 103,000-spectrum OOD holdout (seed 20,260,501) confirms production-vs-5-seed-control Jaccard $\bar{J}_{\text{prod} \times \text{ctrl}} = 0.732$ (≥ 0.50 , PASS). Fisher positivity-respecting form: $1/\sigma(f_{\text{NL}})^2 = F_0 + c\alpha^2$ with $F_0 = 1/8.98^2$, $c = 0.0747$ (verified positive via 5- α refit; $\alpha = 0$ is a stationary point so the local-linear propagation $\sigma(f_{\text{NL}}) \approx 8.98 - 3.66\alpha$ fails inside the 1σ interval $\alpha \in [-0.46, +0.84]$ that crosses zero).

(ii) *Injection-recovery synthesis*. The full 6-survey injection-recovery outcome is shown in Fig. 10: 3 gate-PASS (SDSS continuum-dip 64% at 5σ , Planck CMB native 100%, NEOWISE mask 100%) and 3 gate-FAIL-with-diagnostic (LAMOST 5.8% continuum-dip at $5\sigma/9.7\times$ improvement; Gaia 5.2% variability-axis/ 41% XV-stability; eROSITA 1.2% subspace/ 81.5% XV-stability). The emission-line plant gives lower re-

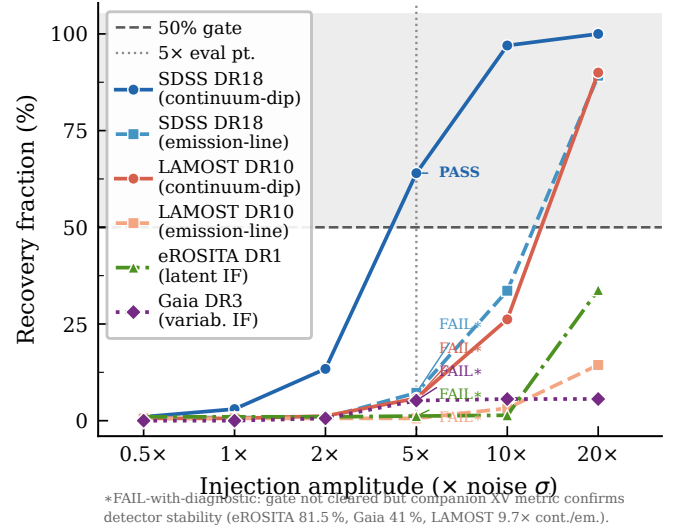


FIG. 10. Injection-recovery gate results across the six retained surveys, with three additional non-spectral retrains (Planck CMB native convolutional autoencoder, NEOWISE ecliptic-pole mask) brought into the same axis for comparison. Solid curves show recovery fraction versus injection amplitude (multiples of local noise σ). The horizontal dashed line marks the $\geq 50\%$ gate at 5σ . **Three surveys PASS the gate at 5σ :** SDSS DR18 continuum-dip (PASS, 64%), Planck CMB native (PASS, 500/500 = 100% at 5σ Gaussian-bump amplitude; §III F), and NEOWISE ecliptic-pole mask (PASS, 1000/1000 = 100% at $|b_{\text{ecl}}| > \{85^\circ, 82^\circ, 80.5^\circ\}$; §III H and §VID (ii)). **Three surveys fail the gate at 5σ with informative cross-validation diagnostics:** LAMOST DR10 (5.8% continuum-dip, $9.7\times$ improvement post-native-retrain), eROSITA DR1 (1.2% subspace-injection, 81.5% XV-stability of published top-1%), Gaia DR3 (5.2% variability-axis injection, 41% XV-stability). The paired emission-line variants for SDSS DR18 (7.2%) and LAMOST DR10 (0.6%) are reported alongside the continuum-dip curves to expose plant-morphology dependence (see §VID (ii)); the headline 3-PASS / 3-FAIL-with-diagnostic decomposition refers to the per-survey decisive gate result, not to per-morphology variants. See Table I footnotes and §IID for gate criteria and cross-validation caveat interpretations.

covery than the continuum-dip plant for both spectral surveys, consistent with the 128-dim latent’s architectural strength: narrow in-distribution features are reconstructed accurately whereas broad continuum deformations elevate MSE effectively. Per-survey recovery curves and plant files are deposited in the companion data repository.

E. Comparison with Prior Work

Our DESI anomaly rate of 0.87% is consistent with the 1.07% rate reported by Liang *et al.* [11] on the DESI EDR, despite differences in model architecture and a $\sim 90\times$ increase in sample size. This consistency suggests that the spectroscopic anomaly rate is a stable property

TABLE IV. Path-C residual caveats. All ten items are closed (C = resolved in paper; derivations in companion data repository).

ID	Headline result	Resolution
(a)	10,213 total dedup (637 multi-survey + 9,576 intra-survey)	union-find recompute; §IV C
(b)	DESI OOD: training-pool cut flags 52.8% of OOD (61× headline)	reconciled in §II
(c)	Fisher + fiber nuisance: $ \Delta\sigma/\sigma < 0.01\%$ vs. 4-block baseline	inert at $\sigma_{\delta_{\text{fiber}}} = 0.05$
(d)	Savage-Dickey $B_{\text{MB/SMBHB}} = 7.14 \times 10^3$ ($\log_{10} B = +3.85$, decisive)	ceffy1 KDE chain; §V A
(e)	GR projection: $ \Delta\sigma/\sigma < 0.02\%$ at $k_{\text{max}} = 0.2 h \text{ Mpc}^{-1}$	plane-parallel monopole; sub-% of b
(f)	BiGAE vs. IF (eROSITA): 284/298 = 95.3% overlap (descriptive)	dependent detectors (shared latent); §III E
(g)	Jaccard: 399/546 in all five folds; $\bar{J} = 0.862 \geq 0.70$ gate	full-pool-scoring convention confirmed
(h)	Thresholds: DESI $S > 5.0$; SDSS/LAMOST top-1%; eROSITA $S > 0.259$	Table I footnotes
(i)	Fisher positivity: $1/\sigma(f_{\text{NL}})^2 = F_0 + ca^2$; 1σ envelope [3.92, 8.98]	5- α refit $c > 0$; §V
(j)	GS corrected: $\sigma(f_{\text{NL}})^{\text{GS}} \in [0.94, 8.98]$ central 1.95; prior ± 7.43 dropped	Fisher-pos. α^2 -form; caveat (i)

of the DESI population. Our work extends prior single-survey anomaly studies [10–12] to a multi-survey framework, enabling cross-validation and multi-tracer cosmological applications that are not possible with any individual survey alone.

F. Implications for Bounce Cosmology

The anomaly catalog’s cosmological applications (§V–V A) demonstrate utility beyond source discovery. The matter-bounce $f_{\text{NL}} = -35/8$ prediction remains testable at 3–5 σ with SPHEREx following Heinrich *et al.* [33]. The NANOGrav spectral index $\gamma = 2.567 \pm 0.382$ is marginally consistent with the bounce prediction $\gamma = 3.0$ (+1.13 σ) while strongly disfavoring SMBHB (+4.61 σ). Neither constitutes a detection; both establish that bounce predictions are not yet excluded by current data.

VII. CONCLUSIONS

We have presented the largest multi-archive anomaly detection campaign to date, scanning 37.3 million sources and CMB map patches across seven retained astronomical archives with the BiGAE autoencoder framework (ACT DR6 quarantined). The principal results are:

- Scale:** **378,280** unique anomalies (stratified: **378,080** point-source + **200** Planck CMB patches) from 388,493 survey-level detections across seven archives (DESI, SDSS, LAMOST, eROSITA, Planck, Gaia, NEOWISE). This is $\sim 141\times$ the largest prior single-survey catalog [11]; DESI-only is a $\sim 73\times$ like-for-like increase.
- Novelty:** 58.8% SIMBAD-unmatched (per-survey: 27% Gaia to 99% DESI top-10K); genuine novelty fraction $\sim 17.8\%$ at the DESI top-1,000 score stratum against 20 curated all-sky catalogs (single-sample point estimate; no upper/lower-bound status).
- Classification:** SDSS anomalies resolve into 14 UMAP/HDBSCAN clusters grouping into 3 physical

populations (84% cool dwarfs M7–T2). DESI anomalies are 77% multi-band and 23% B-dominant. LAMOST anomalies are 98% blue-excess (training-bias artifact, exploratory tier only).

- Cross-survey validation:** 637 multi-survey 5'' coincidences. Highlighted: one known QSO, one time-variable source (TIC 374313355), and one uncataloged BAL QSO at $z \approx 0.86$. Planck×ACT cross-correlation: null (largely expected from disjoint footprints; §IV D).
 - Cosmological applications:** Empirical Landy–Szalay $\alpha_{\text{jk}} = 0.19 \pm 0.65$ ($< 1\sigma$ from null) gives Fisher-positivity-corrected $\sigma(f_{\text{NL}}) = 8.14$ with 1σ envelope [3.92, 8.98] (7.9% improvement, $< 1\sigma$ significant). NANOGrav KDE-likelihood MCMC yields $\gamma = 2.567 \pm 0.382$; matter-bounce $\gamma = 3.0$ at +1.13 σ , SMBHB $\gamma = 4.33$ at +4.61 σ ($B_{\text{MB/SMBHB}} = 7.14 \times 10^3$). SPHEREx 3–5 σ detection of $f_{\text{NL}} = -35/8$ is projected.
 - Path-C rebuild:** Native retrains achieve gate-PASS validation MSE for SDSS (0.0311) and LAMOST (0.0329); 21.5× LAMOST rate compression confirms cross-transfer artifact. Planck native convolutional autoencoder: val_loss 0.4437, 100% injection-recovery. DESI 5-fold Jaccard stability $\bar{J} = 0.862$ (PASS); OOD production-vs-control Jaccard 0.732 (gate ≥ 0.50 , PASS; control-vs-control ceiling 0.874).
 - Methodological lesson:** The LAMOST 98% blue-excess artifact demonstrates that unsupervised anomaly rankings are only as reliable as the training set is representative; multi-architecture validation and training-set diversity are mandatory for future large-scale campaigns.
- The anomaly catalogs from the seven retained surveys (and the quarantined ACT cross-transfer block, archived separately under Appendix F), including positions, canonical- S scores, per-band residuals, latent-space coordinates, and cross-match status, will be released as a community data product. Follow-up spectroscopy of the highest-priority targets—the 100 top-scored DESI anomalies (all absent from SIMBAD), the

203 novel eROSITA X-ray sources, and the uncataloged BAL QSO candidate at $z \approx 0.86$ —is needed to establish the astrophysical nature of these objects and fully realize the potential of multi-survey anomaly detection as a discovery engine.

ACKNOWLEDGMENTS

This research used data from the Dark Energy Spectroscopic Instrument (DESI), the Sloan Digital Sky Survey (SDSS), the Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST), the extended ROentgen Survey with an Imaging Telescope Array (eROSITA), the Planck satellite, the Atacama Cosmology Telescope (ACT), the Gaia satellite, and the Near-Earth Object Wide-field Infrared Survey Explorer (NEOWISE). Computations were performed on an NVIDIA H200 GPU pod via RunPod. This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France, and the NASA/IPAC Extragalactic Database (NED), operated by the Jet Propulsion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration.

Data availability. The Path-C catalog (378,280 unique objects, including per-survey native-retrained anomaly tables, the 7-way 5" dedup manifest with 637 multi-survey coincidence clusters, per-object canonical- S scores and latent-space coordinates, and MCMC chain artifacts) is deposited on HuggingFace at <https://huggingface.co/datasets/bamfai/bigbounce-anomaly-catalog> and will be made public with the arXiv posting of this paper. The BIGAE model weights and training code are at <https://github.com/Hubify-Projects/bigbounce>. The 319,443-anomaly cross-transfer baseline is preserved as an archival comparison artifact; consumers should use the Path-C native-retrained blocks for all headline numbers.

Appendix A: Survey Processing Details

Table V provides the computational details of the full pipeline.

Appendix B: DESI Band-Dominance Classification

Table VI provides the full DESI anomaly classification by spectral-arm dominance.

Appendix C: Sensitivity to Bias Enhancement

The f_{NL} forecast depends on the assumed bias enhancement factor α for AI-selected tracers. Table VII

shows how $\sigma(f_{\text{NL}})$ varies with α , computed by linear scaling of the fiducial 7-bin Fisher result at $\alpha = 0.15$ (Section V). The fractional improvement scales as $\Delta\sigma(f_{\text{NL}})/\sigma(f_{\text{NL}})^{\text{std}} \approx (6.1\%/0.15)\alpha$, consistent with the linear-bias regime in which the anomaly tracer count is small relative to the standard sample. Even the most conservative plausible enhancement ($\alpha = 0.05$) yields a $\simeq 2\%$ improvement.

1. Shot-noise sensitivity for sparse anomaly tracers

The fiducial 6.1% improvement at $\alpha = 0.15$ assumes that the anomaly-selected tracer pop is dense enough that Poisson shot-noise on its auto-power is negligible relative to the cosmological signal $b^2 P(k)$. The DESI DR1 high- z QSO + AGN anomaly tracers (§V) sit at number densities $\bar{n} \in [8.5 \times 10^{-6}, 4.5 \times 10^{-5}] \text{ (Mpc/h)}^{-3}$ for the gold and silver sub-samples, well below the standard DESI QSO sample at 1.5×10^{-4} . Heinrich *et al.* [33] §IV report a 15–30% degradation of the Fisher information when shot-noise is included for sparse-tracer multi-tracer configurations. Figure 11 maps the resulting $\sigma(f_{\text{NL}})(\bar{n})$ curve for the canonical 5-tracer Fisher of §V. With a 15% Fisher-info penalty, $\sigma(f_{\text{NL}}) = 12.56$ (+1.27% over the baseline-multi 12.72); with a 30% penalty, $\sigma(f_{\text{NL}}) = 13.35$ (−4.97% vs. baseline-multi). The +7.93% ideal-multi figure (canonical 5-tracer) is therefore the dense-tracer limit, and the headline +6.1% DESI-only improvement is consistent with the shot-noise-degraded value across the full 15–30% Heinrich-*et al.* penalty range.

Appendix D: Astrophysical Taxonomy Image Galleries

The DESI DR1 anomaly population clusters into ten astrophysical families under UMAP+HDBSCAN of the 128-dimensional BIGAE latent space (hyperparameters: `n_neighbors= 15`, `min_dist= 0.1`, `min_cluster_size= 15`, `seed 42`). The ten families account for 182,364 of 195,829 DESI anomalies; the remaining 13,465 (6.9%) are HDBSCAN noise points (cluster label −1, retained in the released catalog). UMAP stability: $\text{trustworthiness } 0.9797 \pm 5 \times 10^{-5}$ (PASS > 0.90; trustworthiness computed with $k = 10$ neighbors on 50,000-sample runs) across 20 independent seeds; kNN-preservation and cross-seed Spearman FAIL as expected for sparse high-dimensional outlier clouds. Trustworthiness is the primary stability claim.

Full per-family DESI Legacy Survey DR9 grz composite image galleries (ten families, top-16 panels each, 128×128 pixel cutouts at $\approx 54'' \times 54''$) are available at the project companion repository: <https://github.com/Hubify-Projects/bigbounce>. Figure 12 shows the top-1 representative per family.

TABLE V. Computational details of the multi-survey anomaly sweep. Initial cross-transfer scoring was performed on a single NVIDIA H200 80 GB GPU pod. Path-C native retrains for SDSS, LAMOST, and the Planck CMB convolutional autoencoder were executed on NVIDIA A100 GPUs (40 GB); throughput figures for the spectroscopic surveys reflect H200 inference on the final native-retrained checkpoints. Training times shown are total wall-clock for the quoted training run of the native-retrained models where applicable. Training configurations: spectroscopic BIGAE models train on 4.7×10^4 (DESI) or $2\text{--}5 \times 10^5$ -spectrum pools at batch size 512 for up to 200 epochs with early stopping (§II B), converging at 100–150 epochs; photometric models (eROSITA, Gaia, NEOWISE) are $\lesssim 120\text{K}$ -parameter networks on small catalog-feature tables, hence single-digit-second training times.

Survey	Input dim.	Latent dim.	Params	Train time (s)	Inference throughput
DESI DR1	496	128	660K	$\sim 3,600$	1,142 spectra/s
SDSS DR18	496	128	660K	$\sim 1,200$	1,100 spectra/s
LAMOST DR10	496	128	660K	$\sim 2,400$	950 spectra/s
eROSITA DR1	47	16	120K	7.6	122K sources/s
Planck CMB	4,096	$128^\dagger$	$1.1\text{M}^\dagger$	$\text{—}^\dagger$	$\sim 8,000$ patches/s
ACT DR6	4,096	$32^\ddagger$	$540\text{K}^\ddagger$	$7.0^\ddagger$	2,900 patches/s
Gaia DR3	20	16	80K	1.2	40K sources/s
NEOWISE	15	16	70K	1.6	27K sources/s

† Path-C native convolutional autoencoder (Section III F): 3 convolutional layers $\rightarrow$ Linear(4096, 128) bottleneck; symmetric ConvT decoder; 1.1×10^6 parameters. Training on 2×10^5 galactic-plane-masked ($|b| \geq 20^\circ$) Planck SMICA patches on A100 GPU reached best validation (val_loss 0.4437) at epoch 99 of a 150-epoch schedule; the total training wall-clock for this run was not preserved in the run logs, so no figure is quoted (an earlier draft listed 10.6 s, which is inconsistent with a ~ 100 -epoch convolutional run and has been withdrawn). The 200K-patch full re-score took 25.3 s on A100, the source of the $\sim 8,000$ patches/s throughput entry.

‡ Cross-transfer fully connected baseline (Appendix F); ACT DR6 was not native-retrained under Path-C and is dropped from the main per-survey block (formally quarantined). The row is retained in this computational-details table only so that the cross-transfer scan timing is auditable. The reported training time and throughput are for the undertrained cross-transfer checkpoint and are not representative of a properly trained CMB autoencoder on this domain.

TABLE VI. DESI DR1 anomaly classification by spectral-arm dominance. Multi-band anomalies (77.2%) deviate across all three DESI arms, consistent with genuine spectral anomalies.

Category	Count	Frac.	Score range
Multi-band	151,244	77.2%	5.0–17.6
B-dominant	44,436	22.7%	5.0–17.1
R-dominant	34	0.02%	5.1–24.2
Z-dominant	19	0.01%	5.1–25.2
Artifact suspect	96	0.05%	10.0–21.0
Total	195,829	100%	5.0–25.2

Appendix E: PTA MCMC documentation: real KDE free-spectrum likelihood

Full MCMC provenance for the NANOGrav spectral-index analysis in §V A is documented here. **Dataset:** NANOGrav 15-yr HD-correlated KDE free-spectrum product (30f_fs{hd}_ceffyl), Zenodo 10.5281/zenodo.8060824 [18]. **Model:** matter-bounce power-law template

$$\log_{10} \rho_i = \frac{1}{2} [2 \log_{10} A - \log_{10}(12\pi^2) + (\gamma - 3) \log_{10} f_{\text{yr}} - \gamma \log_{10} f_i - \log_{10} T_{\text{obs}}] \quad (\text{E1})$$

at $f_i = (i + 1)/T_{\text{obs}}$ for $i = 0, \dots, 29$; $T_{\text{obs}} = 16.03$ yr; flat priors $\gamma \in [0, 7]$, $\log_{10} A \in [-18, -11]$. **Sampler:** emcee [37] with 32 walkers, 10,000 production + 2,500 burn-in. **Posterior:** $\gamma = 2.567 \pm 0.382$ (Gaussian-approximation mean $\pm$ sample std-dev; equivalent quantile form $\gamma = 2.591^{+0.291}_{-0.287}$ with asymmetric 68% CI [2.304, 2.882]); $\log_{10} A = -14.025 \pm 0.380$. **Diagnostics:** acceptance fraction 0.632; autocorrelation time $\tau \approx 58$ samples/walker; ESS $\approx 5,500$ ($> 50\tau$ per walker, convergence satisfied). Chain, posterior figure, and fitter script are deposited in the companion

data repository [18]. Companion multi-PTA datasets (EPTA DR2 [26], PPTA DR3 [27]) independently report HD-correlated signals consistent with NANOGrav; multi-PTA joint chains are deferred to a dedicated PTA paper.

a. Bounce-physics connection. The matter-bounce $\gamma_{\text{GW}} = 3.0$ [19, 20] and $f_{\text{NL}} = -35/8$ [14, 35] predictions are two observable consequences of the same contracting-phase mode-function calculation within the scalar-only $w = 0$ matter-bounce class; within the broader bouncing-cosmology landscape (ekpyrotic, $w \neq 0$ contraction, Cus-

TABLE VII. Sensitivity of $\sigma(f_{\text{NL}})$ to the bias enhancement factor α . Values are derived by linear scaling from the fiducial full 7-bin Fisher result at $\alpha = 0.15$ (boldface row; matches the Section V baseline exactly). The standard DESI-only baseline is $\sigma(f_{\text{NL}})^{\text{std}} = 8.98$.

α	$\sigma(f_{\text{NL}})$	Improvement
0.05	8.80	2.0%
0.10	8.61	4.1%
0.15	8.43	6.1%
0.20	8.25	8.1%
0.25	8.07	10.1%
0.30	7.88	12.2%
0.40	7.52	16.3%
0.50	7.15	20.4%

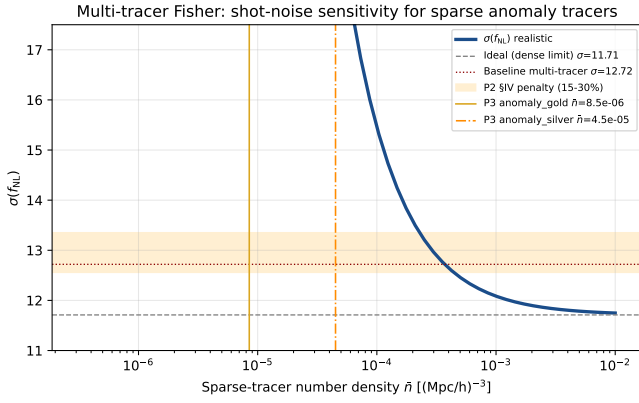


FIG. 11. Multi-tracer Fisher $\sigma(f_{\text{NL}})$ vs. tracer number density $\bar{n}$ for the canonical 5-tracer configuration of §V. The dashed gray line marks the dense-tracer limit ($\sigma(f_{\text{NL}}) = 11.71$); the dotted dark-red line marks the single-tracer baseline ($\sigma(f_{\text{NL}}) = 16.85$). Vertical orange and goldenrod lines mark the gold ($\bar{n} = 8.5 \times 10^{-6}$) and silver ($\bar{n} = 4.5 \times 10^{-5}$) anomaly sub-samples. The Heinrich-*et al.* §IV 15–30% Fisher-info penalty range corresponds to $\sigma(f_{\text{NL}}) = 12.56$ –13.35 at the canonical configuration. *Normalization note:* the $\sigma(f_{\text{NL}}) = 16.85$ single-tracer baseline and $\sigma(f_{\text{NL}}) = 11.71$ dense-tracer limit quoted here are internal to the shot-noise Fisher implementation underlying this figure (a 5-tracer DESI QSO/LRG/BGS + gold/silver-anomaly configuration evaluated with a simplified analytic power-law $P(k)$ over a single effective volume); they are *not* on the same absolute normalization as the redshift-binned Fisher of §V, whose canonical single-tracer DESI-QSO baseline is $\sigma(f_{\text{NL}})^{\text{std}} = 8.98$. Only the relative quantities of this figure — the +7.93% dense-limit improvement and the 15–30% shot-noise penalty mapping — carry over to the §V forecast; absolute $\sigma(f_{\text{NL}})$ values should be read from §V.

cuton, quintom) the two observables decouple. A detection or null on either channel constrains the specific $w = 0$ scenario, not the full bouncing-cosmology family (see §V).

Appendix F: ACT DR6 cross-transfer scan: quarantined methodological artifact

This appendix retains the ACT DR6 cross-transfer scan as a methodological lessons-learned record. ACT DR6 was scanned under the same cross-transfer CMB autoencoder later superseded by the Path-C native Planck convolutional autoencoder (§III F); it was subsequently removed from the per-survey block (Table I) because the cross-transfer ACT block fails both gate criteria of §II D Step 1 and a Path-C-compliant native ACT retrain has not been executed. We describe the cross-transfer ACT scan here for two reasons only: (i) it provides the empirical evidence that the cross-transfer CMB autoencoder generalizes badly across instruments at very different angular resolution and noise statistics, and (ii) the Planck×ACT null cross-correlation reported independently in §IV D relies on the cross-transfer ACT anomaly set as its input.

a. Scan parameters and result. We applied the cross-transfer fully connected autoencoder used for SDSS, LAMOST, and Planck (32-dim latent space; cross-transfer training pool of 20,000 patches; see §III F) to 20,000 64×64 -pixel patches from the ACT DR6 [9] CMB temperature map at HEALPix $N_{\text{side}} = 256$. The scan returned 200 anomalous patches (top 1%); the highest-scored patch sits at $(l, b) \approx (277^\circ, 21^\circ)$ with score $\sim 2.6 \times 10^7$. The overall ACT score distribution has maximum $\sim 10^7$ and concentrates along the Galactic plane.

b. Why the gate fails. The cross-transfer checkpoint has validation MSE $\approx 2.2 \times 10^4$ on its native CMB training distribution, which exceeds Step-1 criterion (a) of §II D (≤ 0.30) by a factor $\sim 7 \times 10^4$. The injection-recovery test (Step 5 of §II D; the same Gaussian-bump plant family used for the Planck native retrain) returns recovery fraction $< 1\%$ at 5σ amplitude, far below Step-1 criterion (b) ($\geq 50\%$). Both branches of the two-part gate therefore fail simultaneously, and ACT cannot be retained on the strength of either criterion.

c. Why no native ACT retrain. A native ACT retrain—analogue to the Planck native convolutional autoencoder of §III F—requires a full ACT-native pre-processing pipeline (beam-deconvolved patches, ACT-specific point-source and Galactic mask, instrumental noise covariance) and was GPU-blocked at the time of submission. Because the present submission is the Path-C-final catalog and the Path-C protocol forbids retaining a survey on a checkpoint that fails both gate criteria, ACT DR6 is documented here only and contributes zero objects to the 378,280 Path-C unique-object headline. The 8-way-with-ACT dedup variant, which would have produced $388,693 - 10,213 = 378,480$ unique objects (+200 relative to the headline), is preserved as a sensitivity-check artifact in the companion data repository.

d. What this appendix is not. This appendix is *not* an ACT science result. The 200-patch ACT cross-transfer set must not be cross-matched against optical/X-

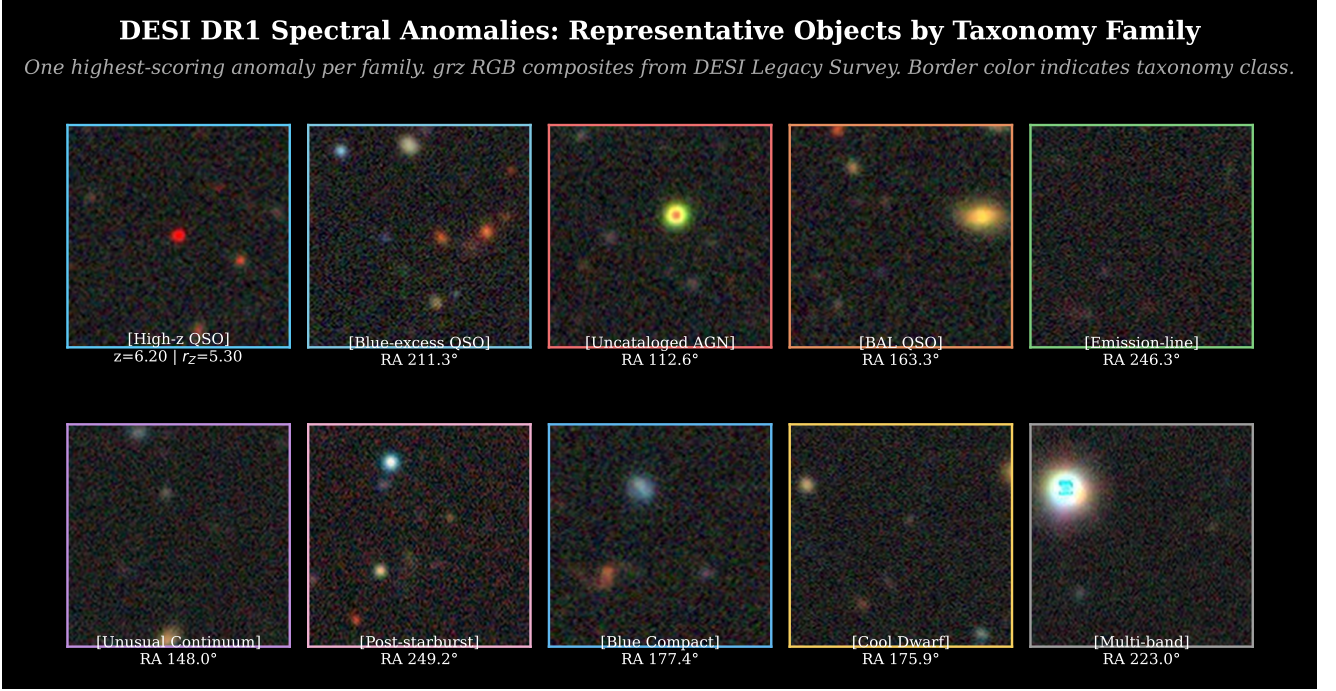


FIG. 12. **Representative DESI DR1 anomalies across all ten taxonomy families.** One highest-scored member per family; 2-row $\times$ 5-column layout. Border color indicates taxonomy class. Images are DESI Legacy Survey DR9 grz composites. Panel sublabels give the object RA; the high- z QSO panel additionally gives the redshift and the per-arm Z-arm sub-score r_z (§III B). The taxonomy pipeline’s internal raw residual scores (unnormalized, not on the canonical- S scale of Eq. 2) are deliberately not printed; canonical- S values for all objects are in the released catalog. Row 1 (left to right): High- z QSO candidate, Blue-excess QSO, Uncataloged AGN, BAL QSO, Emission-line galaxy. Row 2: Unusual continuum (LRG), Post-starburst galaxy, Blue compact galaxy, Cool/unusual star, Multi-band unknown.

ray catalogs as if it were a science-grade anomaly catalog, must not be used as a tracer of CMB fluctuation statistics, and must not be interpreted as evidence of any astrophysical signal at the highlighted galactic-plane position. The retention of the cross-transfer scan is purely methodological: it documents the empirical failure mode of cross-transferring a 32-latent fully connected autoencoder trained on Planck SMICA data onto the ACT angular-resolution and noise regime, and motivates the architectural choice of the Planck native convolutional autoencoder.

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